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Economic Evaluation of Maintenance Strategies for Ground-Mounted Solar Photovoltaic Plants

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Abstract

In this paper we present a generic model which evaluates the economic viability of alternative maintenance strategies for differently sized photovoltaic plants with variable components. We thoroughly review the existing literature on reliability, maintenance and maintenance strategies for PV plants. The model is applied to ground-mounted solar PV plants of three different sizes: 1, 10, and 100 MW_p. The analysis compares immediate corrective maintenance with maintenance strategies of different periodicity (weekly, bi-weekly, and monthly). Simulations take the driving time of the service team and the operation of a spare parts inventory into account. We further analyze the impact of hourly spot prices in contrast to constant prices (feed-in tariffs). We find that corrective maintenance is the most cost-effective strategy. If the service journey takes six hours instead of one hour, corrective maintenance becomes more expensive than weekly periodic maintenance. For a three-hour journey, the size of the PV plant matters. If corrective maintenance is adopted, the establishing of a spare parts inventory should be considered. When volatile hourly spot prices are considered, we find that opportunity costs are lowest (highest) when service teams operate at 8 pm (8 am), irrespective of the day of the week, whereas maintenance events scheduled for Wednesdays at 8 am or 2 pm are, given current EPEX spot prices, the most cost-effective in terms of total costs.

Keywords: Photovoltaics, Corrective maintenance, Germany

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List of Abbreviations

AC	Alternating Current
BOS	Balance of System
CBM	Condition-Based Maintenance
CDF	Cumulative Density Function
CM	Corrective Maintenance
DC	Direct Current
EPEX	European Stock Exchange
MC	Monte Carlo
MPPT	Maximum Power Point Tracking
MTBF	Mean Time Between Failure
O&M	Operations and Maintenance
PDF	Probability Density Function
PM	Periodic Maintenance
PV	Photovoltaics
PV-RPM	Photovoltaic Reliability and Performance Model
QMO	Quantitative Maintenance Optimization
SNL	Sandia National Laboratory
TRY	Test Reference Year
UB ₉₅	Upper Bound of the 95% Confidence Interval

List of Symbols

c_{dr}	Costs per man-hour of driving
c_w	Costs per man-hour of work
C_f	Fixed costs
$CF(t)$	Cash flow in period t
$E(G(t), T_u(t))$	Electricity production per module in period t
$f(t)$	Probability density function at time t
$F(t)$	Cumulative density function at time t
$G(t)$	Solar irradiance in period t
$h(t)$	Hazard rate function at time t
i	Discount rate
$n_{ar}(t)$	Number of offline arrays in period t

n_{arrays}	Number of arrays
n_{mod}	Number of modules per array
$p_{el}(t)$	Sales price of electricity in period t
Pr	Probability
P_{STC}	Nominal power of a solar module
$R(t)$	Reliability function at time t
$R_x(25a)$	Probability that the component x will not fail within 25 years
t	Time
t_{dr}	Time of the journey to the PV plant
$t_{n,offline}$	Offline hours of array n
t_{repl}	Time consumption for the replacement of an item
$t_{lifetime}$	Lifetime of the simulated PV plant
T_a	Ambient temperature
T_c	Solar cell temperature
T_{NOCT}	Normal operating cell temperature
T_{STC}	Cell temperature under standard test conditions
$T_u(t)$	Temperature in period t
V	Present value
β	Shape parameter of the Weibull function
β_P	Solar cell power temperature coefficient
η	Scale parameter of the Weibull function
λ	Failure rate

1 Introduction

Growing environmental concerns have led to a change in the energy markets worldwide. Ambitions to reduce fossil fuel consumption and greenhouse gas emissions led to record additions for the second consecutive year for wind and solar photovoltaic (PV) energy in 2015. The share of renewable energy sources had increased to 23.7% of global electricity production by the end of 2015, with wind and PV accounting for 77% of new installations. The total capacity of PV installations worldwide has now reached 227 GW, 40 GW of which is in Germany (REN21, 2016). German government programs and subsidies have helped to integrate PV power plants into today's energy market. Through technical progress and scale and learning effects, the investment costs of PV plants have been reduced by 75% since 2006 (Wirth, 2016). Additional

cost reductions in production in combination with incentives offered by the government will further increase the installed capacity of photovoltaic in the foreseeable future (Zhang et al., 2013). With increasing numbers of PV plants and decreasing investment costs, the operating costs of PV plants are of growing importance. Stakeholders are interested in low operation and maintenance (O&M) costs, while at the same time they rely on the ability of the PV plant to produce the expected amounts of electricity in order to secure the invested capital. The selection of a maintenance strategy is therefore a process that needs to be evaluated in the light of technical and economic conditions.

The best suited maintenance strategy for a PV plant depends on multiple factors. PV plant size, component configurations, location, global irradiation, etc. are important criteria to factor in when selecting a strategy. Nowadays, most PV plants are operated with a corrective maintenance strategy in combination with preventive maintenance carried out according to the component manufacturer's guidelines. These preventive measures are limited to the useful life period of a component, since electronic components are not subject to the same deterioration as mechanical components, and the anticipation of failures is more difficult (Jardine and Tsang, 2013). With increasing size of the PV plants, the number of components also rises. Avoiding any failures at all is a very unlikely scenario. In the case of a failure, corrective maintenance is usually carried out right away in order to achieve a high availability of the PV plant even though a strategy that leads to high availability might not always be coherent with the most cost-effective strategy.

The objective of this paper is to develop a model which supports decision makers when selecting a cost-effective maintenance strategy. The model simulates a PV plant and its components over a lifetime of 25 years. Components are selected and failures simulated with respect to available failure data in the literature. The focus of this paper is on the economic evaluation of possible actions following any failure of components. Corrective maintenance which is carried out right after the detection of a failure is compared to periodically scheduled maintenance events as a form of deferred corrective maintenance. Resulting O&M costs and opportunity costs stemming from lost revenues are evaluated and compared. Restrictions arise from the limited availability of failure data in the literature, but the model is implemented in a flexible way so that the number and characteristics of the components can be adjusted when more reliable data become available. The model is an event-based simulation which is implemented in MATLAB. The failure times for the components are randomly selected from their probability distributions, and the resulting O&M and opportunity costs are hence also random. Monte Carlo

simulation is used to produce probability distributions of the results. This way, all possible outcomes and their probabilities can be evaluated and discussed.

Section 2 provides a state-of-the-art literature review on reliability, maintenance as well as maintenance strategies for PV plants. Additionally, the theoretical background for reliability modeling, maintenance strategies, and maintenance optimization is set out. Section 3 describes the model developed and its sub-models for PV plant reliability, the service team, and the economic evaluation. The selected input parameters for the model are described and discussed in section 4. The resulting histograms are discussed and maintenance strategies evaluated with respect to the mean values, median, and the upper bound of the 95% confidence interval in section 5. Section 6 concludes and gives an outlook on possible further investigations.

2 Literature Review

After an introduction to maintenance in sections 2.1 and maintenance optimization in section 2.2, existing literature on photovoltaic reliability and maintenance is reviewed in section 2.3. An overview of PV power plants is then given in section 2.4, followed by the theoretical background of reliability modeling (section 2.5), maintenance classification (section 2.6), and maintenance optimization (section 2.7).

2.1 Introduction to Maintenance

Prior to World War II, the requirements on production were relatively low, so that industrial plants were solely maintained on a breakdown or *corrective* basis. From the 1950s onwards, marketplaces became more competitive, labor wages rose, and more machinery was used in the production industry. Tolerance for plant downtime decreased, and production demanded better maintenance, which led to the development of scheduled *preventive* maintenance: Components and machineries were assumed to more likely fail with increasing age and were therefore overhauled on a regular basis. In the 1960s the aviation industry began to question the current maintenance strategies and realized that the replacement of components on a scheduled basis leaves a lot of the potential untapped. Through investigations they discovered that the time of failure of a component does not have to be time-dependent but is often condition-dependent. This led to the introduction of *condition-based* maintenance (Brown and Sondalini, undated). Maintenance is executed when deterioration indicators exceed predefined thresholds.

2.2 Literature on Maintenance Optimization

Maintenance optimization includes qualitative and quantitative methods to evaluate different maintenance strategies under predetermined conditions. Dekker (1996) gives an extensive summary of applications of maintenance optimization from 1982 to 1996. Vatn et al. (1996) present an approach for optimal maintenance scheduling in the production industry. Important factors that are considered in the optimization process are: safety and health concerns, environmental protection, maintenance costs and costs of lost production. Aven and Dekker (1997) introduced a general framework for the optimization of replacement times. The authors considered a number of different models, including block and age replacement models. More recent case studies for maintenance optimization are reviewed in Sharma et al. (2011). Examples of maintenance optimization can also be found in the energy industry. Several studies investigated the reliability and maintenance of both on- and offshore wind farms (Andrawus et al., 2007; Andrawus, 2008; Besnard, 2013; Igba et al., 2015; Kerres et al., 2015). A structured asset management model including a reliability block diagram is developed by Andrawus (2008). Besnard (2013) proposes decision models for cost efficient maintenance for offshore wind farms and Kerres et al. (2015) develop a stochastic model for assessing the life-cycle cost and availability of wind turbines resulting from different maintenance scenarios. Sinha and Steel (2015) use failure analysis as a basis for designing a condition based prognostic maintenance plan for wind turbines. They use Failure Modes Effects and Criticality Analysis as well as the concept of Risk Priority Numbers. An investigation into condition based maintenance for marine renewable energies, i.e. offshore wind, tidal flow and wave energy, has been conducted by Mérigaud and Ringwood (2016). De Faria Jr. et al. (2015) present a review of predictive maintenance methods and methodologies for fault diagnosis based on gas analysis related to transformer maintenance. Further investigations have been conducted on the reliability of nuclear power plants and power distribution grids. Nilsson (2009) presents models based on the methods of Life Cycle Cost and mathematical optimization, applied to wind and nuclear power plants. Hilber (2008) examines the optimum balance of preventive and corrective maintenance for power distribution networks by solving the multi objective optimization problem with meta heuristics.

2.3 Photovoltaic Reliability and Maintenance Studies

The reliability and maintenance of photovoltaic (PV) power plants is important to the stakeholders of a PV plant, since any failure of components in combination with inadequate maintenance strategies can lead to great financial losses (Zhang et al., 2013). The maintenance of PV plants has attracted more attention in recent years due to the rapid increment of photovoltaic

power plants. Other industries have used maintenance strategies and optimization before, and nowadays these are seen as value adding activities rather than a necessary expensive evil (Sharma, 2011).

There are studies that have evaluated the reliability and availability of PV systems and their components by using general reliability analysis methods. Such methods include reliability block diagrams, fault tree analysis and Markov models. In a reliability block diagram, a system and its components are represented in a graphical illustration to assess the systems overall reliability. The blocks representing the systems components are linked depending on their effects on the system. Fault tree analysis is a top down analysis in which the reliability of a system is analyzed using Boolean logic to combine multiple lower-level events. Markov models consist of a set of states in which a system may be. The transition to another state occurs randomly and does not depend on the states the system has been in before (Birolini 2010; Čepin, 2011; Grinstead & Snell, 1997). Some papers focus on reliability (subsection 2.2.1), whereas others use reliability models and methods of maintenance optimization to estimate or minimize the effect of component failure on the economics of PV power plants (subsection 2.2.2).

2.3.1 Studies Focusing on PV Reliability Issues

There are several studies that focus on the reliability of PV system components: Mishra and Joshi (1995) use a reliability estimation method to predict the probability of success of various components of PV systems, including diodes, circuit breakers, capacitors, and more. Inverter reliability is investigated in grid-connected PV systems using the design-of-experiments technique of Chan and Calleja (2009). Harb and Balog (2013) introduce a stress-factor reliability method to calculate the mean time between failure of a PV module-integrated inverter via a usage model approach. Dhople et al. (2012) study the reliability of a DC/DC converter by using Markov analysis. Inverter and diode reliability are investigated by Li (2012) with a condition-dependent risk assessment. The reliability of large arrays in different configurations is investigated under the application of probability theory (Gautam and Kauschika, 2002). Some papers also examine entire PV systems: fault tree analysis and probability analysis are used to assess the reliability equation of a PV system in order to improve the performance of diagnosis in the case of failure (Zini et al., 2011). Ahadi et al. (2014) also use fault tree analysis to assess the total component reliability and overall reliability of a PV system. The Monte Carlo simulation technique is applied and a MATLAB program developed for a reliability analysis of a small isolated PV power system (Moharil and Kulkarni, 2012). The reliability of a PV system is estimated with a hierarchical reliability block diagram by Collins et al. (2010a) and Collins et al (2010b). Caldwell et al. (2016) use reliability modeling and accelerated life testing to formulate

a cost model of photovoltaic systems with different architectures. The failure rates available in the literature are very limited, because multiple studies draw on the same primary sources for their applied data. Table A1 gives an overview of the available failure data for selected components (see Appendix).

2.3.2 Studies on PV Reliability and Effects on Economics

First investigations of PV plant reliability and economics can be found in Stember et al. (1982), who developed a model for studying the effects of reliability issues on the economics of PV plants. Because of the lack of failure data, hypothetical reliability data were used as input. Inverter reliability was integrated into a PV energy yield estimation by Dhople et al. (2010). Lubin (2012) examined the effects of component failure on the economics of a PV plant by developing a simulation model and using actual failure data from a 17 kW PV installation.

In recent years studies have emerged that consider maintenance actions in their investigations on the reliability of PV systems. Multiple studies were conducted at the Sandia National Laboratory (SNL) in Albuquerque, New Mexico, USA, with the intention of developing a photovoltaic reliability and performance model (PV-RPM) (SNL, 05.07.2016). As part of this program, Collins et al. (2010a) and Collins et al. (2010b) investigate the reliability and availability of a fielded PV system and conduct a sensitivity study. The application of the PV-RPM is demonstrated in Miller et al. (2012) and is used to compare three different PV system designs. The PV-RPM is still under development but will include the option to choose different maintenance types once finished (SNL, 05.07.2016). Further research is conducted by Dhople et al. (2012), who use the Markov method to estimate PV system reliability and performance under different maintenance regimes. The Markov method is also used to examine a stand-alone PV system including repairs (Theristis and Papazoglou, 2014). Reliability and energy yields of a small grid-connected PV system are investigated and the effect of maintenance examined by Perdue et al. (2015). They evaluate different scheduled maintenance regimes but do not estimate emerging operations and management (O&M) costs. Polo et al. (2015) try to incorporate monitoring data into dynamic maintenance tasks. A generic model for PV power plants of variable size and with variable components that incorporates and evaluates maintenance strategies is yet to be developed.

2.4 Photovoltaic Power Plants

A PV power plant can be subdivided into the PV generator and the balance of system (BOS) components. The generator consists of PV *cells* that are combined with PV *panels* which convert the sunlight into electricity. The panels can be arranged in series to form a *string*. Multiple strings connected in parallel form an *array* (Fig. 1).

Connecting PV panels in series increases the voltage, whereas a parallel configuration increases the resulting current. Different configurations of the panels are possible and they directly affect the topology and structure of the BOS components (Cai et al., 2015). Common setups are the centralized, the string, and the modular configuration. The BOS components include the racking structure, DC/DC converter, inverter, switches, DC and AC protection devices, wiring, and others (Cai et al., 2015). One of the main tasks of the BOS is to convert the DC power generated by the panels into higher voltage AC power which can be fed into the electricity grid.

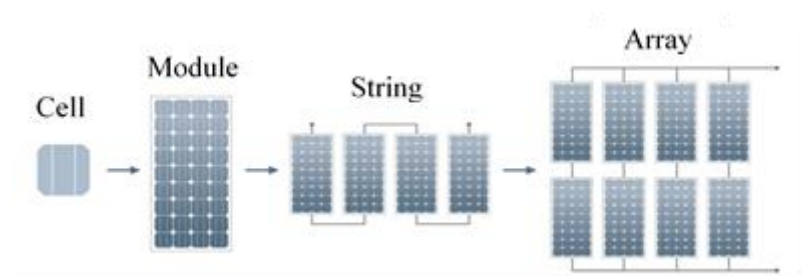


Fig. 1: Photovoltaic cell, module, string, and array

Source: Stapleton et al. (2013)

Different topologies for photovoltaic system are discussed in the literature and are applied in actual power plants. Large-scale grid-connected PV power plants are usually set up in a centralized- or string-inverter configuration (Li, 2012). In a centralized set up, multiple strings are connected to the same inverter whereas each string has its own inverter in the string configuration. An advantage of a string inverter is individual maximum power point tracking (MPPT) for each string as well as easier fault detection in the case of failures. (Zhang et al., 2012). Another possible topology is the micro-inverter configuration. Micro-inverters are integrated into the PV panels as one electrical device. By using micro-inverters, a high modularity and enhanced safety can be achieved (Li, 2012). MPPT can be applied for each module individually. This can be particularly beneficial under partial shading conditions (Zhang et al., 2013).

2.5 Reliability Modeling

Reliability is defined as “the ability of an item to perform its required function without failure, under given environmental and operational conditions and for a stated period of time” (ISO 9000). In reliability modeling, the time of failure of an item is a random variable which can be described by various probability functions (subsection 2.5.1). Common failure distributions will be elaborated in subsection 2.4.2.

2.5.1 Probability Functions

The *probability density function* (PDF) $f(t)$ describes the relative likelihood of a random variable taking on a given value, e.g. the time of failure. The probability of a failure occurring before some specified time t can be expressed by the *cumulative density function* (CDF) $F(t)$. It relates to its PDF by:

$$F(t) = \int_0^t f(t)dt. \quad (1)$$

The exponential PDF and the CDF are depicted in Fig. 2. The probability that a system will not fail within the time interval is given by the *reliability function*:

$$R(t) = 1 - F(t) = \int_t^{\infty} f(t)dt. \quad (2)$$

It is sometimes called the survival function, since it describes the probability that the system will survive at least until time t . The *hazard rate function* $h(t)$ defines the probability of failure in an infinitesimally small time period between t and $t + dt$ given that the subject has survived up till time t (Birolini, 2010; Jardine and Tsang, 2013):

$$h(t) = \frac{f(t)}{R(t)}. \quad (3)$$

The *failure rate* λ describes the frequency with which a component fails. It is expressed in failures per unit of time and its inverse value represents the *mean time between failure* (MTBF) for a component.

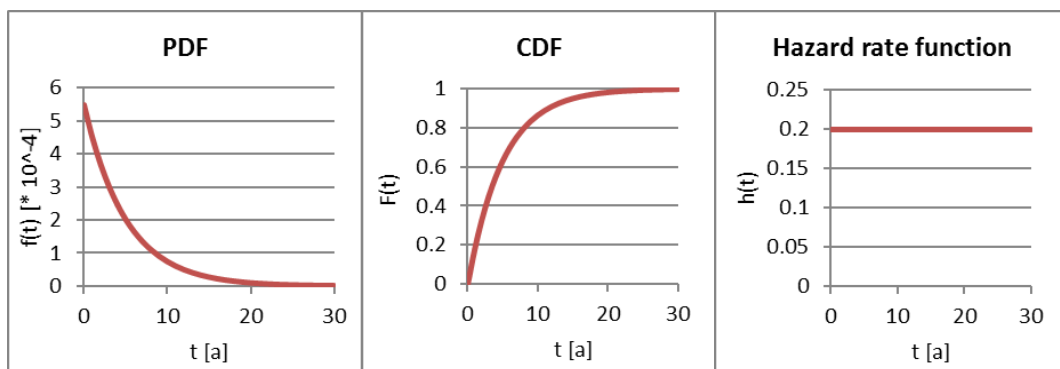


Fig. 2: Exponential probability density function (PDF), cumulative density function (CDF), and hazard rate function

2.5.2 Probability Distributions

The failure characteristics of components are different from each other and can be described by various probability distributions. Common PDFs used in reliability modeling include the exponential and the Weibull distribution. The former is found to be typical of complex industrial plants and many electronic components (Jardine and Tsang, 2013). Its density function is defined as

$$f(t) = \lambda * \exp(-\lambda t) \quad \text{for } t \geq 0, \quad (4)$$

where λ is the failure rate. A characteristic feature of the exponential PDF is its constant hazard rate function, which implies that the probability of failure is the same at any time t if the component has not failed before. The Weibull distribution is one of the most frequently used PDFs, because it is very versatile and can be fitted to a large number of failure characteristics (Briolini, 2010). The density function of the two parameter Weibull distribution is

$$f(t) = \frac{\beta}{\eta} \left(\frac{t}{\eta}\right)^{\beta-1} \exp\left[-\left(\frac{t}{\eta}\right)^{\beta}\right] \quad \text{for } t \geq 0, \quad (5)$$

where β is the shape parameter, and η is the scale parameter. If β is equal to 1, the two-parameter Weibull distribution is equivalent to the exponential distribution, which means that failures occur independently over time (Jardine and Tsang, 2013). A β value lower than 1 signals a decreasing failure rate over time whereas β values greater than 1 imply an increasing failure rate. These three phases are shown in the bathtub curve, which represents a common failure pattern of an item's lifetime (Fig. 3). It starts with the *infant mortality* ($\beta < 1$) which includes initial failures caused by inappropriate handling or problems in manufacturing, followed by independently occurring failures over time in the *useful life* period ($\beta = 1$) which are unpredictable and cannot be avoided. The bathtub curve ends with the *wear out* period ($\beta > 1$) which comprises of typical end of life failures (Tont et al., 2014). The bathtub curve in Fig. 3 is simply for illustrative purposes; the exact shape of the distribution of each component depends on multiple factors, such as mechanical or electrical component, manufacturing, quality, system layout, and site-dependent and environmental stresses (Perdue et al., 2015).

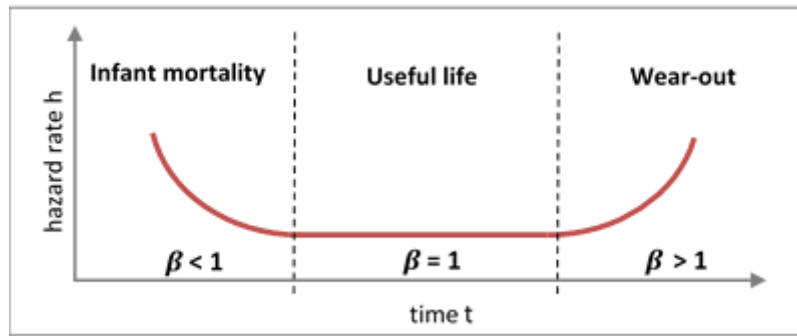


Fig. 3: Illustrative bathtub curve of the hazard rate function

2.6 Maintenance Classification

Maintenance is defined as the “combination of all technical, administrative and managerial actions during the lifecycle of an item intended to retain it in, or restore it to, a state in which it can perform the required function” (DIN EN 13306). The maintenance strategy describes the way in which these actions are executed and combined to achieve the maintenance objectives. Possible objectives are the increase of availability, minimization of total cost, improvement of product quality, environment preservation, or enhanced safety (Birolini, 2010; DIN EN 13306). Maintenance divides into preventive and corrective maintenance (Fig. 4). The objective of the former is to retain the functionality of an item and thereby avoid failure beforehand, whereas corrective maintenance takes place after the occurrence of a failure and reestablishes the item’s functionality.

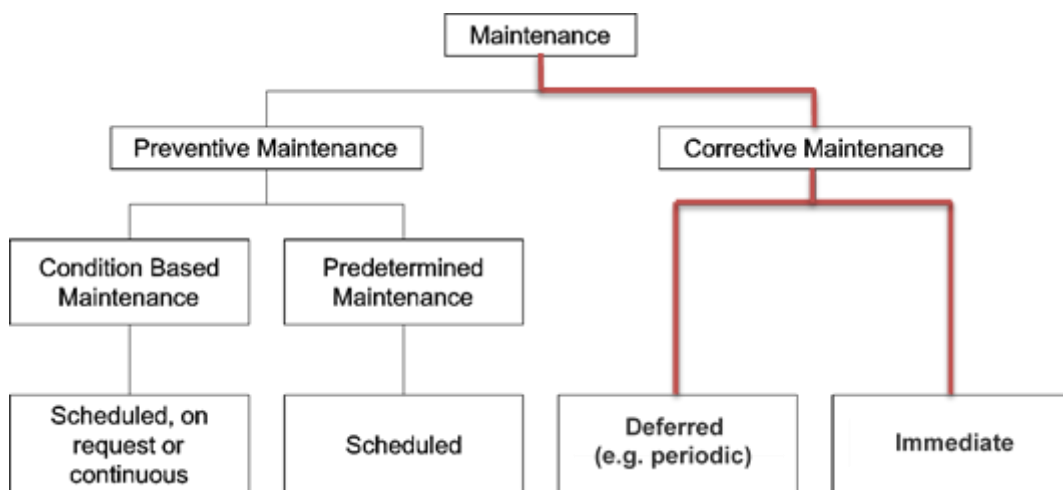


Fig. 4: Maintenance classification

Source: DIN EN 13306, modified

2.6.1 Corrective Maintenance

Corrective maintenance is initiated after fault recognition. The corrective action can take place immediately or it can be deferred in accordance with given rules. Periodically scheduled maintenance is a form of deferred corrective maintenance, since the item's functionality will not be restored immediately but the next time that a maintenance event is scheduled (DIN EN 13306). This strategy helps to avoid unnecessary inspections and repairs of the system and the resulting costs and downtimes (Kerres et al., 2015).

2.6.2 Preventive Maintenance

Preventive maintenance is carried out to reduce the risk of failure or the degradation of the functionality of an item. It can be divided in predetermined maintenance and condition-based maintenance (CBM). Predetermined preventive maintenance can be performed in fixed intervals of calendar or operational time or according to another usage parameter (Froger et al., 2015). Preventive CBM is executed when deterioration indicators exceed predefined thresholds. A prerequisite is the ability to monitor the component condition (Kerres et al., 2015). Enough knowledge about the failure characteristics of the components is also required to be able to anticipate failures.

2.7 Optimization Theory

Maintenance optimization is “a process that attempts to balance the maintenance requirements (legislative, economic, technical, etc.) and the resources used to carry out the maintenance program (people, spares, consumables, equipment, facilities, etc.)” (System Reliability Center, 2003). The resulting maintenance strategy has to be reevaluated on a regular basis since the input variables, such as interest rate, component cost, failure characteristics, etc., may change over time. Maintenance optimization is therefore a continuous process (Andrawus, 2008).

There is qualitative and quantitative maintenance optimization. The former includes methodologies like reliability-centered maintenance and total productive maintenance. Scarf (1997) and Arthur (2005) noted that qualitative maintenance optimization is often clouded with subjective opinion and experience. They further recommend quantitative maintenance optimization (QMO) (Andrawus, 2008). QMO includes various deterministic and stochastic models in which both the cost and benefits of maintenance are quantified, and an optimum balance between both is obtained (Dekker, 1996; Sharma et al., 2011). QMO methods include the Markov method, Petri Nets, Monte Carlo (MC) simulation, and more (Li, 2012). Computer simulations, such as the MC simulation, are more flexible than other QMO methods and are preferred for complex systems and more complicated problems (Zhang et al., 2013; Stember et al., 1982). The MC

method as a tool of maintenance optimization relies on repeated random sampling of failure times for the considered components of a system. Since the input values are random numbers, the resulting values are random as well. The repetition of the same simulation provides the decision maker with a range of possible outcomes and the probabilities of these outcomes occurring. A probability distribution of the results is the basis for further evaluation.

3 Model Specification

The model developed estimates the effects of different maintenance strategies on photovoltaic power plants' O&M costs as well as production losses because of system downtime. It is based on a model developed by Kerres et al. (2015), who did an economic evaluation of maintenance strategies for wind turbines. Similarly to that work, our model consists of two submodels: (1) A reliability model to simulate a PV power plant and to assess its availability and (2) a model for the service team that reacts according to a predetermined maintenance strategy. The models are explained in detail in the following sections. The O&M costs as well as the production losses depend on the random failure times of the components and are hence also random variables. Monte Carlo simulation will be used to estimate the probability distribution of the results.

3.1 Reliability

The reliability model is set up in a flexible way, and the number and attributes of the components can easily be modified. The model simulates a PV farm over a predetermined number of years and records the offline hours for each string/array. Whether a string or an array is the subject of investigation depends on the components and parameters that are chosen. In the further context, the word *array* may be substituted by the word *string*. The arrays themselves are modeled as a series connection of multiple components. If one of these components fails, the array is offline until a maintenance event has occurred. The other parallel running arrays are not affected. Additional components connect the arrays to the electrical grid. Since these components are connected to all arrays, the failure of one of them sets all arrays offline. Fig. 5 demonstrates a block chain diagram of the model proposed. The components deteriorate through a binary process, in which each component can have two states: a 'good' state in which it fulfills its required function and a 'failed' state, in which it does not. The transition from state 'good' to state 'failed' is described by a random variable T . Instances of T are assumed to be identically distributed and independent. T is determined with respect to the components' failure characteristics, which can be an exponential or a Weibull probability density function.

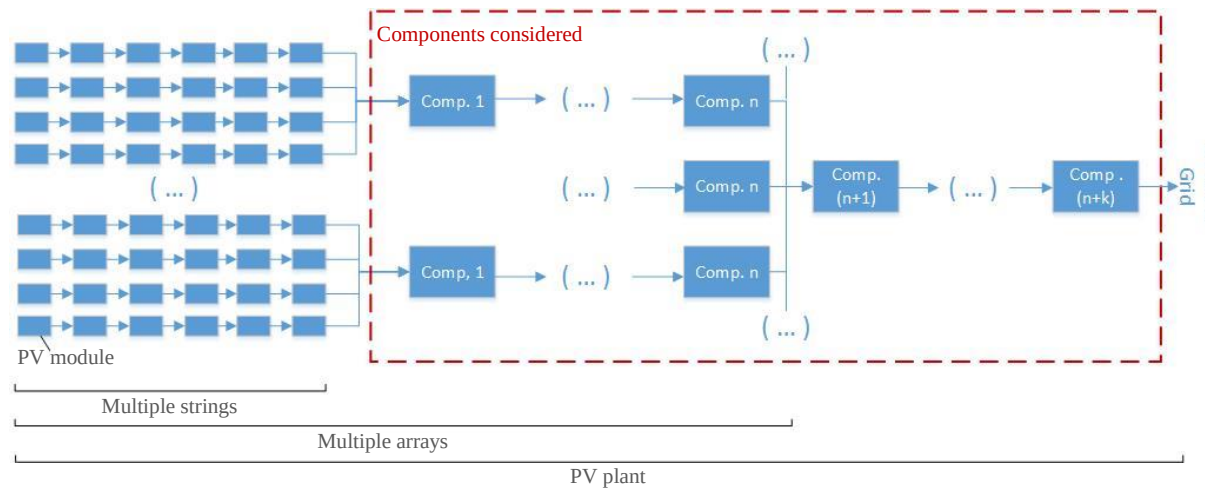


Fig. 5: Block chain diagram of a generic PV system with n components per array and k components connecting the arrays to the grid

3.2 Service Team

The service team can be assigned different tasks on site. These tasks can be combined arbitrarily and the total costs equal the sum of all the underlying basic actions. The actions are listed in Table 2. In this table, c_w is the costs per man-hour of work, c_{dr} the costs per man-hour of driving, and C_f denotes the fixed costs associated with the renewal of the component in question. All variables except for c_w and c_{dr} are component-specific. The following basic actions are possible: (1) *Drive*: Maintenance staff drive to or from the power plant. Action has no influence on the power plant. The costs depend on the time to get to the plant; (2) *Switch on/off*: The array changes state. No costs associated with this action but the downtime of the array will lead to production losses. In the case of a component failure, the array shuts itself down automatically and needs to be restarted manually. This is only possible if no component is in the ‘failed’ state; (3) *Order*: Certain components might have to be ordered and are thus not immediately available. Small components are usually assumed to have no lead time and can be installed right away; and (4) *Replace*: The component is replaced by a new one with the deterioration state ‘good’ and an age of zero. The array is turned off during the exchange, and costs result from the working time of the service team and the procurement of the new component.

Table 2: Possible service team actions

Action	Effect	Time consumption	Array available	Cost
Drive	Drive to/from power plant	t_{dr}	yes	$2t_{dr}c_{dr}$
Switch on/off	Array changes state	0	-	0
Order	Item available after lead time	0	-	0
Replace	Item renewed	t_{repl}	no	$2t_{repl}c_w + C_f$

3.3 Economic Evaluation

In order to evaluate the trade-off between system downtime and maintenance costs, it is necessary to estimate lost power sales. The loss in energy sales due to offline arrays can be expressed as

$$CF(t) = n_{ar}(t) n_{mod} E(G(t), T_u(t)) p_{el}(t), \quad (6)$$

where $n_{ar}(t)$ is the number of offline arrays in period t , n_{mod} is the number of modules per array, $E(G(t), T_u(t))$ marks the electricity production per module which is subject to the solar irradiance G and the temperature T_u in period t , and $p_{el}(t)$ is the sales price of electricity in period t . Since the cash flows occur at different times, a present value approach is used to make them comparable:

$$V = \frac{CF(t)}{(1+i)^t}, \quad (7)$$

where V is the present value, $CF(t)$ is the cash flow in period t , and i is the discount rate per period (Nilsson, 2009). The power production per module is calculated for every hour of the year by (Hellmann et al., 2014):

$$P = \frac{G}{1000 \frac{W}{m^2}} P_{STC} (1 - \beta_P (T_c - T_{STC})), \quad (8)$$

where P is the power production per module, G is the solar irradiance, P_{STC} is the nominal power of the solar module, β_P is the solar cell power temperature coefficient, T_c is the solar cell temperature, and T_{STC} denotes the temperature under standard test conditions. The solar cell temperature is determined as (Hellmann et al., 2014):

$$T_c = T_a + (T_{NOCT} - 20^\circ C) \frac{G}{800 \frac{W}{m^2}}, \quad (9)$$

where T_a is the ambient temperature and T_{NOCT} is the normal operating cell temperature.

3.4 Implementation

The model is implemented in MATLAB, and the simulation over the lifetime of the PV plant is event-structured. A flow diagram of the model is presented in Fig. 6. The simulation starts at $t_{start} = 0$ and is finished at $t_{end} = 8760 * t_{lifetime}$, where t is the lifetime of the PV plant in years. The unit of the time t is *hours*.

An event can be a failure event, a maintenance event, or the end of the simulation. Once the simulation is started, a first failure time for every component will be randomly determined from the components' probability distribution. The occurrences of the maintenance events depend

on the chosen maintenance scenario. Scheduled maintenance events will be added to the event list in the initiation process, whereas corrective maintenance events will not be added until a failure event occurs. Periodic maintenance events are only executed if one or more components are in the failed state. All events are stored and ordered in a file by time of occurrence. As the simulation clock advances, the events will be selected sequentially from the event list.

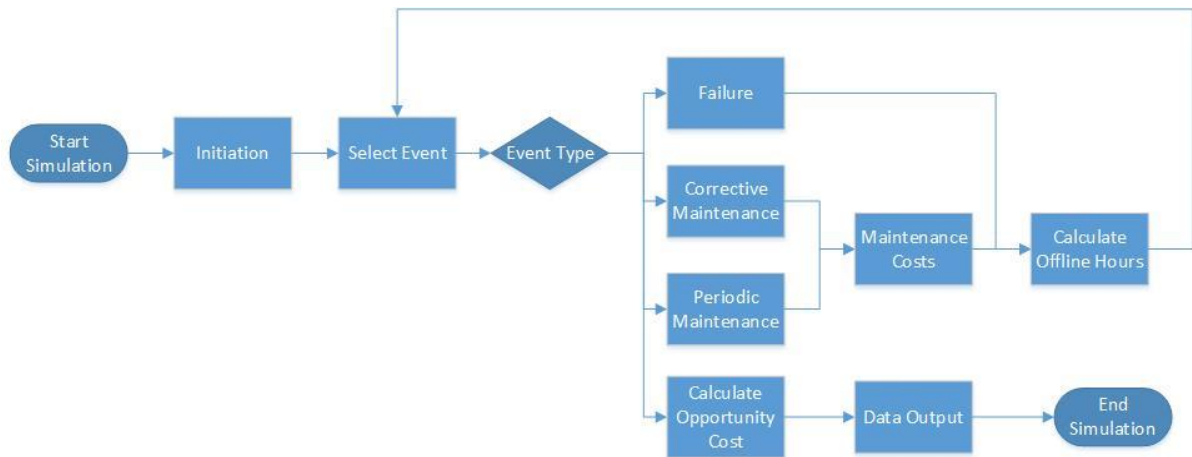


Fig. 6: Flow diagram of the proposed model

Once the initiation is finished, the simulation clock starts running. If the event is a component failure, the affected component state will be set to ‘failed’ and the corresponding array will be marked as ‘offline’. In the case of the corrective maintenance strategy, a new maintenance event for corrective action will be added to the event list. A corrective or periodic maintenance event will send the service team to the PV plant. After the failed components have been repaired, the component state will be reset to ‘good’. The corresponding array is turned back on and the hours that have passed since the component failed will be marked as ‘offline’. All actions that had to be conducted by the service team which incur costs are added to the O&M bill.

Once an event has been processed and the data recorded, the simulation clock moves on to the next event time, and the procedure is repeated until $t = t_{end}$. At the end of the simulation the opportunity cost will be calculated. All hours of an array’s lifetime are either marked as ‘online’ or ‘offline’. The lost revenue is then calculated as stated in section 3.3. The O&M costs, and the opportunity costs stemming from forgone revenue and the availability, are then stored in an external file for further evaluation.

4 Model Parametrization

The model is implemented in such a way that different PV system configurations and different types of failure data can be used for the simulation. The objective of this study is to present the validity of the model developed with a generic PV system setup. The investigations will focus on the comparison between corrective maintenance which is carried out right away and periodically scheduled maintenance events as a form of deferred corrective maintenance. Sections 4.1 and 4.2 present the chosen components and economic parameters for the simulation. The applied maintenance scenarios are defined in section 4.3.

4.1 Component Selection

In this paper a central inverter topology is investigated. The component selection was based on the following criteria: (1) Availability of failure data, (2) Compatibility with the model developed, (3) Severity of negative consequences on power production in the case of failure, and (4) Results and insights from previous reliability studies.

A review of the literature revealed that failure data for the components of photovoltaic power plants are scarce. Many different studies draw on the same sources for the failure data used in their investigations. The applied failure data for the selected components is listed in Table 3. The available failure data as well as the selected data have an exponential probability density function. This implies that a component's failure rate is constant over the entire lifetime (see subsection 2.5.2). The selected components have to be compatible with the model developed. A PV module for example does not fit the criteria, since its failure would usually not lead to the outage of an entire array or the entire power plant. Previous studies on the reliability of PV plants ranked the components of a PV plant with different methodologies (Zini et al., 2011; Colli, 2015; Ahadi et al., 2014). The insights gained from these papers were also used as a basis for the selection. The following components were selected as components of an array: (1) DC Switch, (2) Circuit Breaker, and (3) Inverter. The arrays are connected to the grid through these components: (1) AC Switch and (2) Circuit Breaker. Fig. 7 shows the system configuration. All components are assumed to be non-repairable, which means a replacement of the whole component is necessary after a failure in order to restore functionality.

Table 3: Selected failure rates for the components

Component	Failure rate [10^{-6} Failure/h]	Reference
DC Switch	0.696	Power Systems Committee (1997)
Circuit Breaker	5.712	US Department of Defense (1991)
Inverter	12.9	Cai et al. (2014)
AC Switch	0.696	Power System Committee (1997)

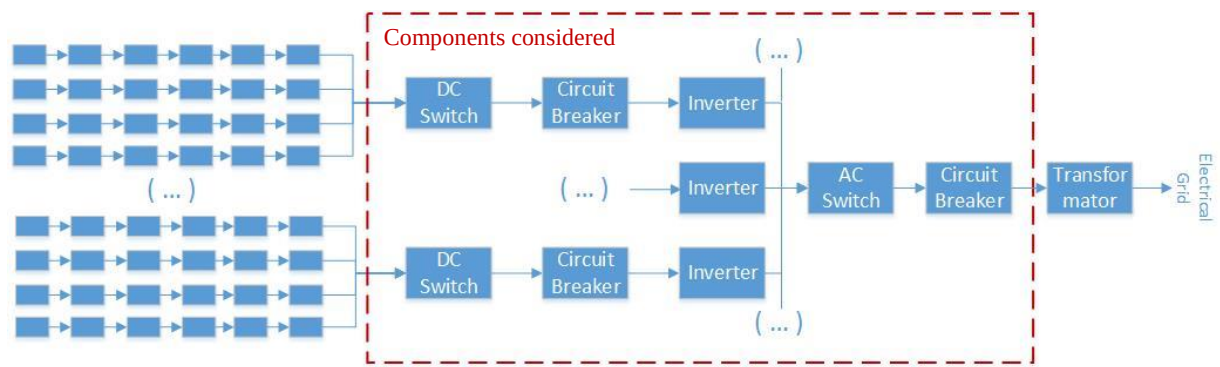


Fig. 7: PV system configuration with selected components

4.2 Economic Parameters

The economic parameters include all parameters that are associated with the service team and its operations and the calculation of the lost revenues. Subsection 4.2.1 defines the parameters for service team and spare parts, whereas subsection 4.2.2 explains the calculation of the power production.

4.2.1 Service Team and Spare Parts

Table 4 shows the component prices that are due when a component is renewed. The prices shown are assumptions that are based on prices in unpublished maintenance contracts (kindly provided by Trianel GmbH). Equipment cost are presumed to be included in the component cost. An inverter is assumed to be available after 12 hours, a switch after 6 hours, and the circuit breakers are assumed to be available in a spare parts inventory and therefore have no lead time. Installation time for all components is assumed to be 8 hours.

Table 4: Selected parameters for the components

Component	Investment costs C_f [€]	Lead time [h]
Inverter	8000	12
Circuit Breaker	400	0
DC Switch	4200	6
AC Switch	4200	6

All maintenance actions are performed in teams of two technicians and they are charged on an hourly basis. A man-hour of work is priced at €100 and a man-hour of driving at €75. Driving time to and from the power plant is assumed to be one hour if not stated otherwise.

4.2.2 Power Production

The following assumptions for the array configuration are made: (1) each array consists of 120 strings, (2) each string consists of 24 PV modules, and (3) each module has 260 W_p, resulting

in a total of 6.24 kW_p per string and 748.8 kW_p per array. As a result, the number of arrays determines the power output of a PV plant. The PV system setups under investigation are displayed in Table 5.

Table 5: PV systems under investigation

Power class [MW]	Number of arrays		Actual power output [MW]
	Theoretical	Assumed	
1	1.34	1	0.749
10	13.35	13	9.73
100	133.55	134	100.34

Weather data are provided by the German Federal Institute for Research on Building, Urban Affairs and Spatial Development ("Bundesinstitut für Bau-, Stadt- und Raumforschung - BBSR"). They provide "test reference years" (TRY) which include climate data on an hourly basis for several locations in Germany. Further information on the calculation of the TRY can be found on the BBSR website (BBSR, 12.07.2016). Scenarios are conducted with the TRY data for Bremerhaven, Germany and Schwäbische Alb, Germany. The yearly sum of global irradiation is about 980 kWh/m² and 1150 kWh/m², respectively. In Germany the sum of the global radiation varies between 950 and 1260 kWh/m² (DWD, 15.07). If not stated otherwise the TRY for Bremerhaven are used for the simulations. The electricity price is fixed at €89.4 per MWh, which is an actual value achieved in a call for bids for photovoltaic plants in Germany in 2015. The interest rate for the calculation of the net present value is assumed to be 5%.

4.3 Maintenance Scenarios

The maintenance scenario determines what actions the service team will take in different situations. Multiple maintenance scenarios are investigated (Table 6).

Table 6: Maintenance scenarios

Maintenance type	Frequency	Waiting time for available service team [h]	Additional component lead time
Immediate	-	1-24	Yes
Deferred (periodic)	weekly	0	No
Deferred (periodic)	biweekly	0	No
Deferred (periodic)	monthly	0	No

In the *corrective maintenance scenario*, the service team will react whenever a failure occurs. It is assumed that the service team is available after a certain waiting time. This waiting time is

a random number between 1 and 24 hours. Not all spare parts are available immediately. If the new component has a lead time, the service team has to wait until the component in question is available for installation.

In the *periodic maintenance scenario*, regular maintenance events are scheduled. In the case of a failure, the service team will not react right away, but install the new component in the next scheduled maintenance event. All defective components are renewed when the service team is on site. Since the events are scheduled there is no additional waiting time for the service team and no additional lead time for the components. Multiple frequencies will be investigated for periodic maintenance, i.e. weekly, biweekly, monthly.

5 Results and Discussion

Four key indicators are used to evaluate the results of each maintenance scenario: (1) Unavailability, (2) Operations and maintenance costs, (3) Opportunity costs, and (4) Total costs. Unavailability is defined as:

$$\frac{\sum_1^n t_{n,offline}}{t_{lifetime} * n_{arrays}}, \quad (10)$$

where n_{arrays} is the number of arrays, $t_{n,offline}$ is the sum of offline hours of array n , and $t_{lifetime}$ is the lifetime of the PV plant. The O&M costs are determined by the present value of all maintenance actions. This includes costs associated with the service team and costs stemming from the acquisition of new components. The opportunity costs are the present value of all the lost electricity revenues incurred by offline arrays. It is closely related to the unavailability because the unavailability is one of the factors that affect the calculation of the lost production. The total costs are the sum of the opportunity and the O&M costs.

In section 5.1 the resulting probability distributions of the key indicators are discussed and compared with each other. Further evaluation is conducted in section 5.2 by evaluating the mean values and the median of the resulting probability distributions. Additionally, the upper bound of the 95% confidence interval (UB₉₅) is investigated to evaluate risks associated with the costs.

5.1 Probability Distributions

The probability distributions are based on a Monte Carlo simulation that has been realized with 10,000 runs per scenario. The distribution for the unavailability and the opportunity cost are discussed in subsection 5.1.1 since the unavailability forms the basis for the calculation of the opportunity cost. O&M and total costs are evaluated in subsection 5.2.2.

5.1.1 Unavailability and Opportunity Costs

The probability distribution for the unavailability of the 1, 10, and 100 MW_p classes with the weekly scheduled maintenance scenario is depicted in Fig. 8, plots (a)-(c), respectively. Fig. 8(a) shows a normal distribution with a lower bound of zero percent. This is plausible since the 1 MW_p PV plant consists of only one array and is hence just a setup of five components in series. The lower bound of 0% unavailability represents the probability of no component failures over the lifetime of 25 years. This probability can be calculated using eqs. (1) and (2):

$$Pr = R_{DCS}(25a) * R_{CB}(25a) * R_{Inv}(25a) * R_{ACS}(25a) * R_{CB}(25a) \cong 0.36\%, \quad (11)$$

where $R_x(25a)$ is the probability that the component x will not fail within 25 years. Fig. 8(c) is a double-peaked distribution. The peaks at 0.2% and 0.3% are accounted for by failures in the components that connect the arrays to the grid. A failure in one of those components leads to the outage of the entire plant and therefore to a higher unavailability. The peaks are more extreme for larger PV plants because there are more arrays in parallel. The probability distribution for the 10 MW_p plant is slightly right-skewed, similarly to the distribution of the 100 MW_p plant, but peaks cannot be identified because there are only 13 instead of 134 arrays in parallel.

Figure 9 shows the probability distribution for the unavailability of a 100 MW_p PV plant with different maintenance strategies. All strategies again lead to a double-peaked distribution, as seen before in Fig. 8(c). The execution of corrective maintenance right after the occurrence of a failure leads to an unavailability between 0.06% and 0.12% (Fig. 9(a)). If weekly periodic maintenance is applied, the center of the distribution shifts towards higher unavailability, and the spread increases (Fig. 9(b)). These developments become more extreme with higher intervals, i.e. 2 weeks and 1 month (cf. Fig. 9, plots (c) and (d)). The time until a component has been renewed can be of any length up to 1 month in the monthly maintenance strategy, which

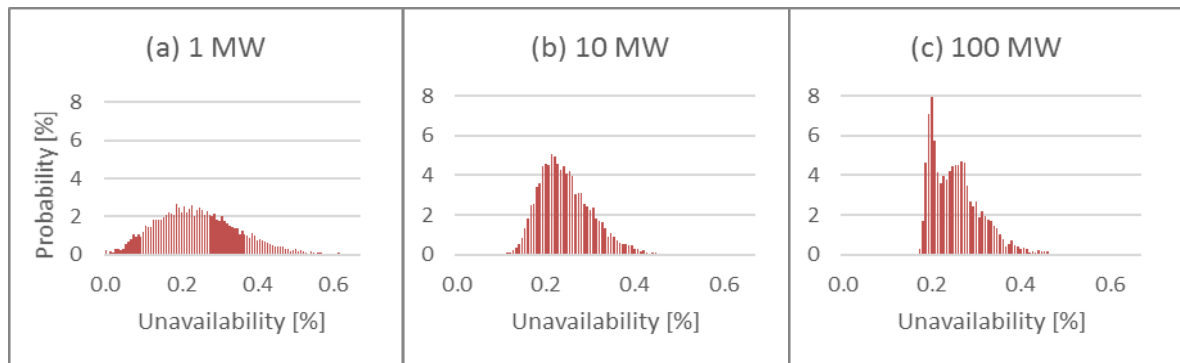


Fig. 8: Probability distribution of the unavailability for different power classes in the weekly scheduled maintenance scenario

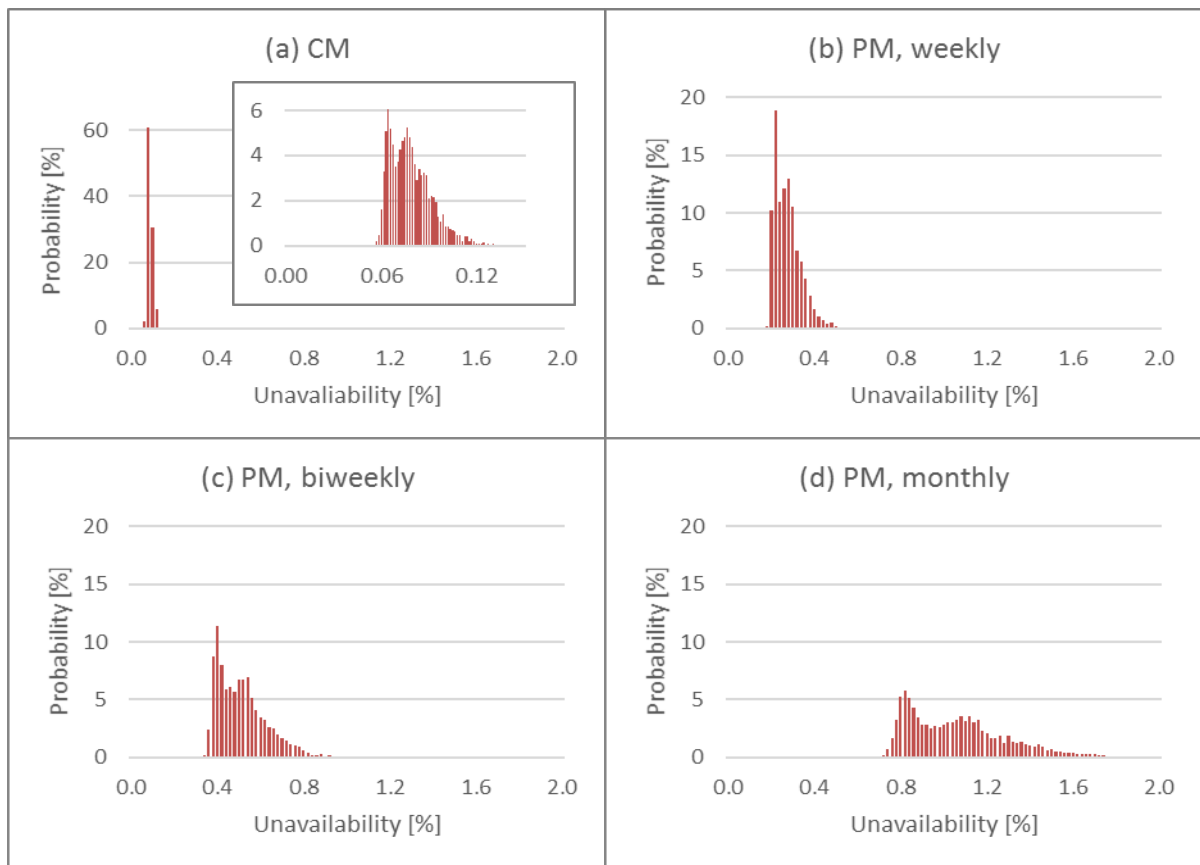


Fig. 9: Probability distribution of the unavailability for different maintenance strategies (corrective maintenance (plot (a); Inlay depicts the same results with smaller histogram intervals), weekly periodic maintenance (plot (b)), biweekly periodic maintenance (plot (c)) and monthly periodic maintenance (plot (d)), 100 MW plant

explains the wider spread compared to the other strategies. Notice that these trends are the same for the 1 and 10 MW PV plants (Appendix, Figs. A1 and A2) and similar observations have been made by Perdue et al. (2015).

The opportunity costs for the periodic maintenance strategies in two weeks and one month intervals are represented in Fig. 10, plots (a) and (b). A decrease in the maintenance frequency leads to a shift towards higher unavailability and a wider spread. This is the same trend as seen before in the unavailability because the unavailability is one of the factors that plays into the calculation of the opportunity costs. The different shapes for the probability distributions of the unavailability and the opportunity costs are due to the discounting and the solar irradiation. Unavailability does not necessarily lead to lost revenues when there is no solar irradiation at that time.

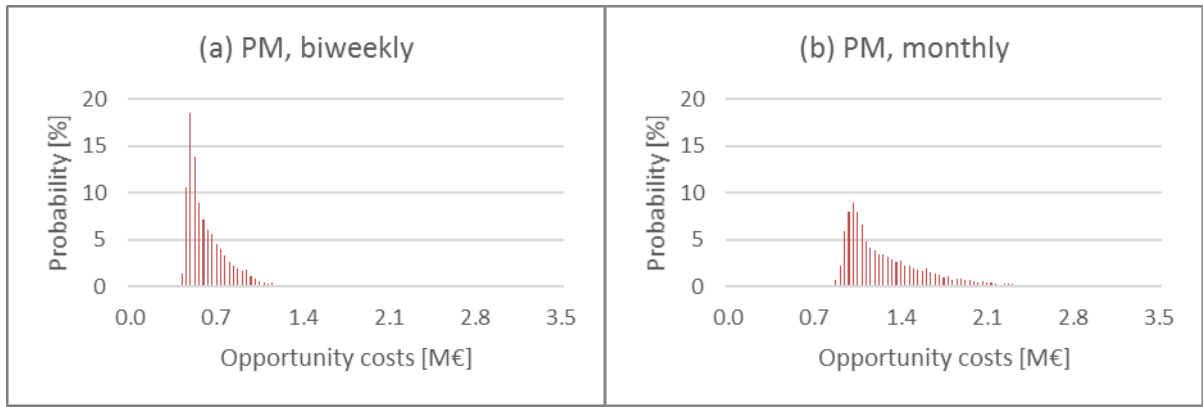


Fig. 10: Probability distributions of the opportunity costs for different maintenance strategies, 100 MW_p plant

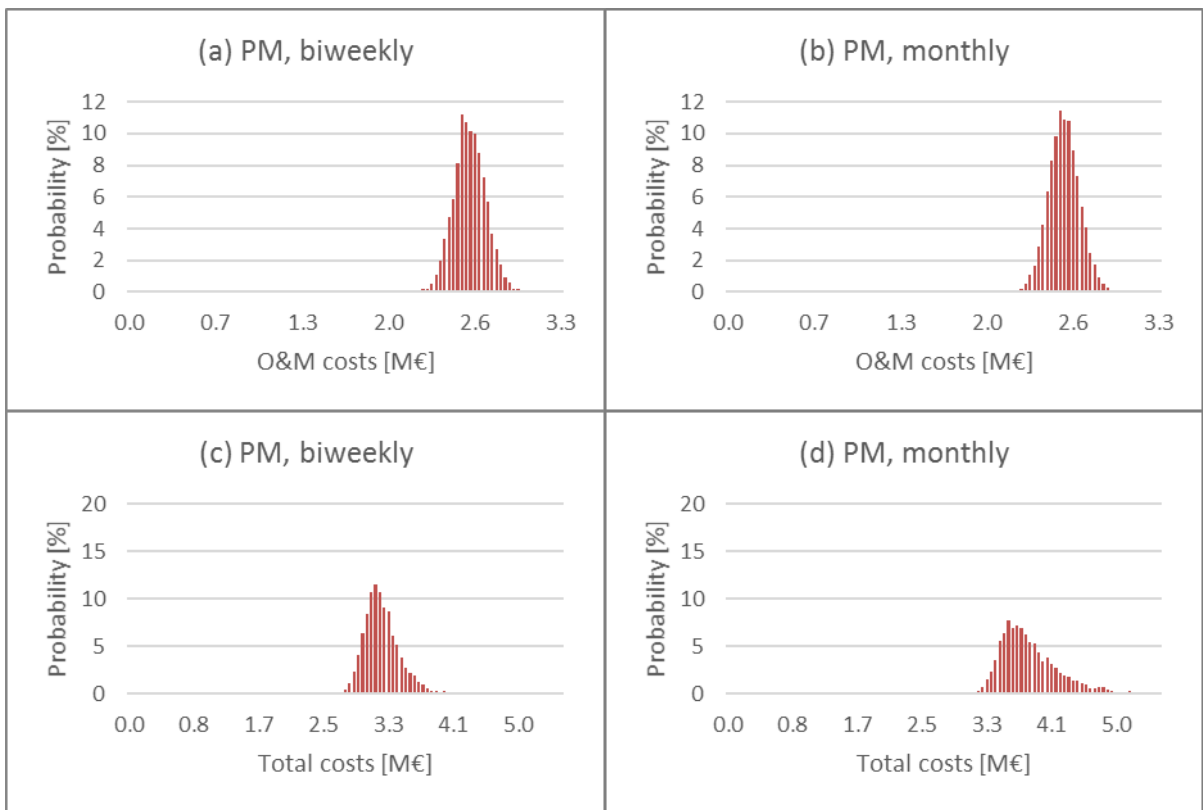


Fig. 11: Probability distributions of the O&M and total costs for different maintenance strategies, 100 MW_p plant

5.1.2 O&M Costs and Total Costs

The O&M as well as the total costs for the periodic maintenance strategies in two weeks and one month intervals are shown in Fig. 11. The O&M costs depicted in Fig. 11, plots (a) and (b) have no significant differences. O&M costs stem from the service team operations and the acquisition of spare parts. Since the distributions are similar to each other but there are fewer

service team operations in the monthly periodic maintenance scenario than in the other scenarios, it can be assumed that the main part of the O&M costs stem from the acquisition of spare parts. The total costs are the sum of the opportunity and the O&M costs. The total costs shown in Fig. 11, plots (c) and (d) are the results of the O&M costs in Fig. 11, plots (a) and (b) as well as the opportunity costs reported in Fig. 10, plots (a) and (b), respectively.

5.2 Mean Value and Median

In order to compare the maintenance strategies, the probability distributions are evaluated using three key figures: the mean value, the standard deviation, and the median. For all four key indicators (unavailability, opportunity costs, O&M costs, and total costs) a low mean value as well as a low median is preferred. A small standard deviation implies lower risks of deviation from the expected mean value. For this reason, the upper bound of the total costs that can be expected with 95% confidence (UB_{95}), similar to the value at risk concept in investment evaluations, is investigated. A risk-averse PV plant operator prefers a lower total cost upper bound, because it indicates a lower risk of high losses.

The base scenarios with the parameters elaborated in section 4 are discussed in subsection 5.2.1. In the following, the driving time of the service team to the PV plant is altered (subsection 5.2.2), the effect of a spare parts inventory and a more reliable inverter is investigated (subsections 5.2.3 and 5.2.4), and some further inquiries are made with hourly electricity spot prices (subsection 5.2.5).

5.2.1 Base Scenarios

The mean value of the unavailability for different maintenance strategies and plant sizes is depicted in Fig. 12. All maintenance strategies have a nearly constant unavailability for the three PV plant sizes. A slight increase can be noticed for the monthly periodic maintenance strategy. The 100 MW_p PV plant consists of 134 arrays and therefore of 404 system components altogether. All these components combined have a failure rate of 0.002594 failures per hour or a mean time between failures (MTBF) of 16 days, whereas the 41 components of the 10 MW_p plant have an MTBF of 162 days (see Appendix, Fig. A3). Since up to one month can pass in the monthly maintenance scenario before the next service team operation is executed, a higher unavailability for the 100 MW_p plant is plausible. Comparing the strategies among each other shows an increase in unavailability with lower maintenance frequency of the service team because the corrective action is deferred according to the chosen strategy.

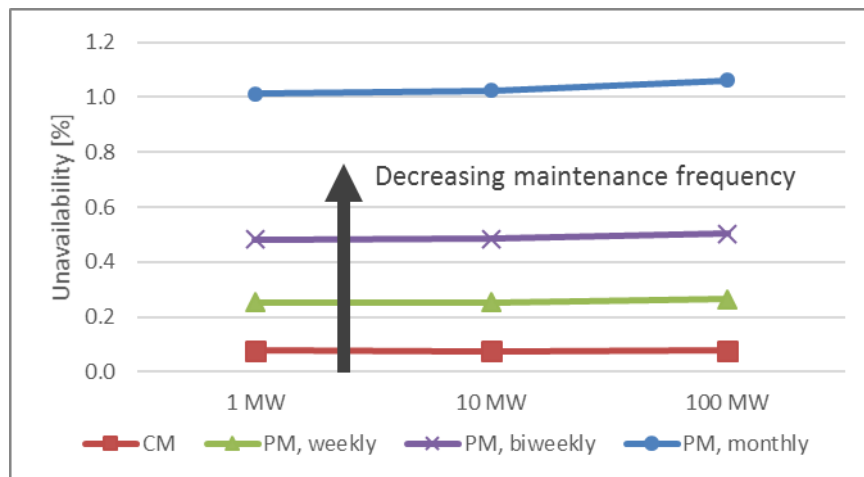


Fig. 12: Mean value of the unavailability for different maintenance strategies and plant sizes

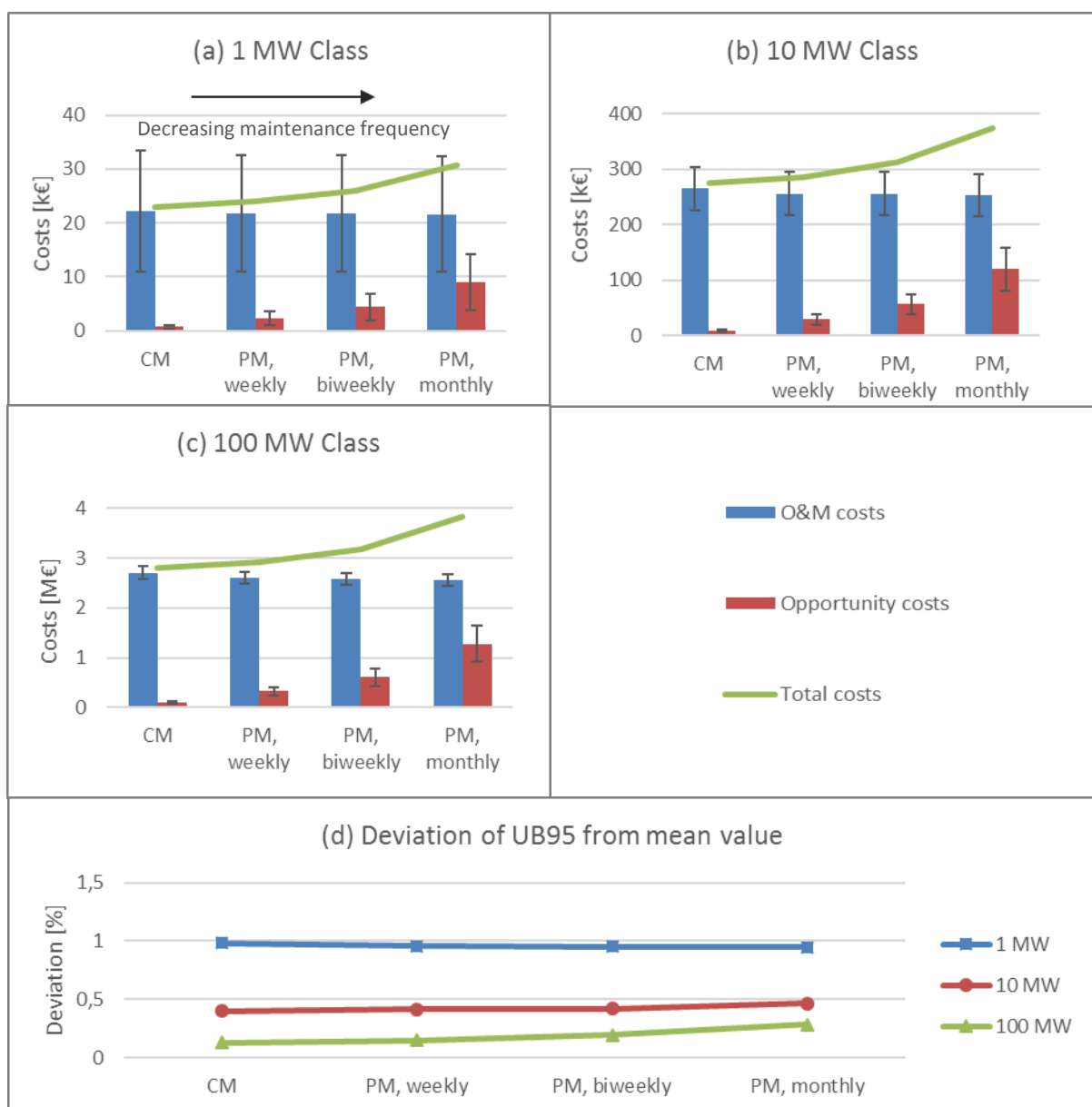


Fig. 13: Cost structure of a 1 (plot (a)), 10 (plot (b)) and 100 (plot (c)) MW_p PV plant with different maintenance scenarios. (d) Deviation of the upper bound of the 95% confidence interval from the mean value of the total costs

Fig. 13 plots (a)-(c) present the cost structure, comprising the opportunity costs, O&M costs, and the total costs of the three PV plant power classes under different maintenance regimes. A decrease in the maintenance frequency in all three cases leads to an increase in the total costs. This is due to the opportunity costs which rise when the plant is unavailable for a longer time because of fewer maintenance events. These fewer maintenance events also lead to a slight decrease in the O&M costs, but not enough to compensate for the lost revenues.

This trend is the same for all investigated PV plant sizes. The standard deviation of the O&M and the opportunity costs is smaller for higher power classes. This is also validated by the deviation of the upper bound of the 95% confidence interval (UB_{95}) from the mean value in Fig. 13(d). The deviation is higher for smaller power classes. Note that a smaller standard deviation and a smaller UB_{95} can be advantageous for the risk management.

The specific O&M, opportunity costs, and total costs in € per kilowatt peak and year are shown in Fig. 14 plots (a)-(c), respectively. The specific O&M costs of all maintenance scenarios are subject to the scale effect and decrease with ascending PV plant size. While all of the periodic maintenance strategies are associated with similar specific O&M costs, the corrective maintenance strategy is the most expensive one. This is due to the corrective action that is initiated right after every failure, whereas multiple components can be renewed at once in the periodic maintenance scenario. Another reason is that potentially two service team operations are necessary for one renewal in case the first inspection reveals that a component has to be replaced but is not available for immediate installation. The specific opportunity costs shown in Fig. 14(b) lead to the same conclusion that has been drawn before in Fig. 13, i.e. that fewer maintenance events lead to higher opportunity costs. The specific total costs depicted in Fig. 14(c) are also subject to scale effects – they decrease with ascending plant size – but the most expensive strategy is the one with the lowest maintenance frequency.

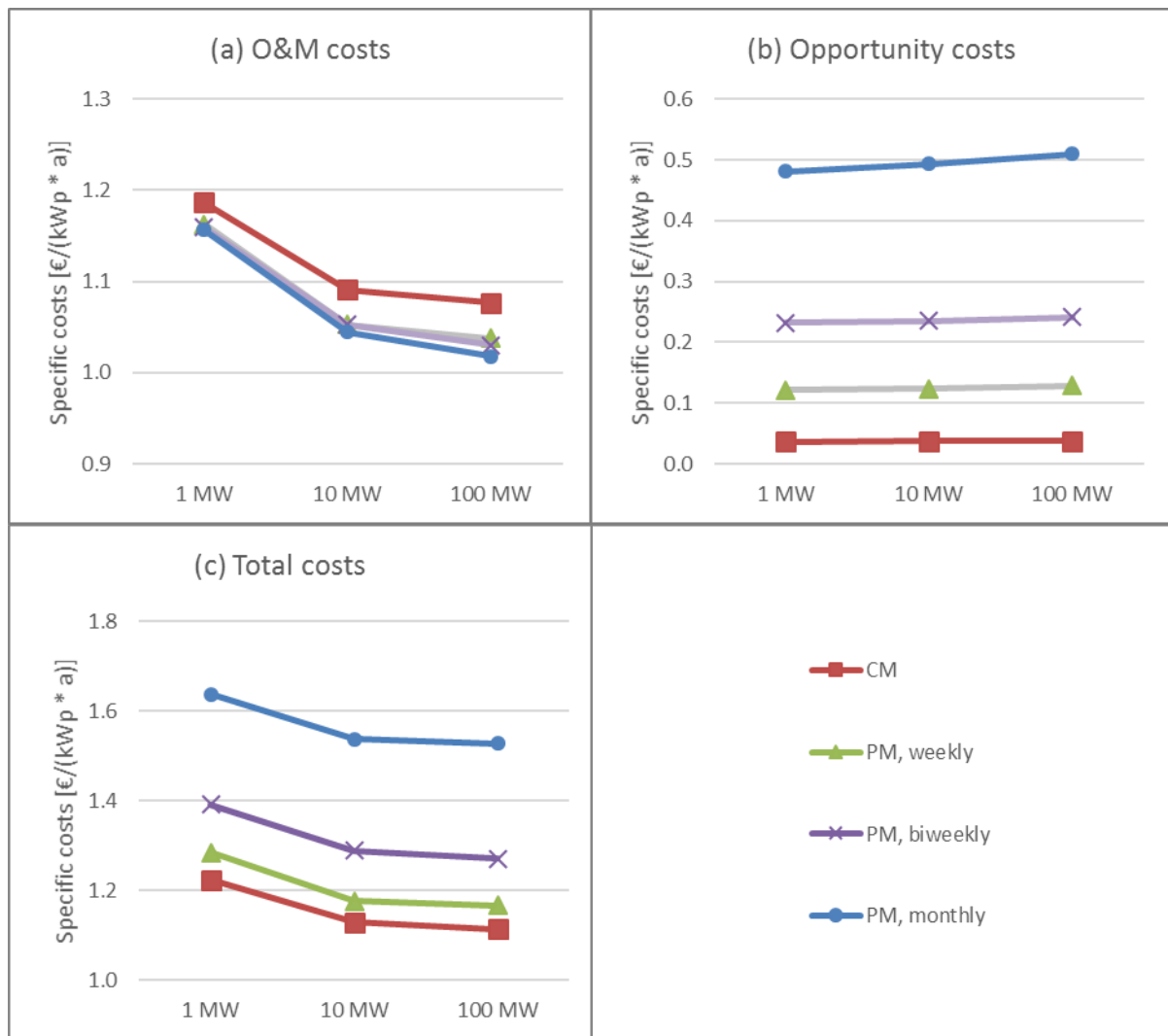


Fig. 14: (a) O&M, (b) opportunity, and (c) total costs per kW_p and year for different maintenance regimes and power classes

5.2.2 Variation of Driving Time

The results reported above show that a periodic maintenance strategy with decreasing maintenance frequency leads to an increase in opportunity costs and an under-proportional decrease in O&M costs (Fig. 13). When the driving time to the PV plant increases, the total O&M costs rise because each service team operation becomes more expensive with distance. It remains to be investigated whether the costs avoided in service team operation through periodic maintenance can actually overcompensate the opportunity costs thereby incurred.

Fig. 15 displays the total costs that emerge for all four maintenance strategies under the assumption that the service team's driving time is 1, 3, or 6 hours one way. For all strategies and PV plant classes the total costs rise with a longer journey for the service team. This is plausible because the number of failures and therefore the number of service team operations

is the same within each maintenance scenario, but every journey is three or six times as expensive due to the longer journeys. For a driving time of one hour, corrective maintenance is the most cost-effective choice, regardless of plant size. Under the assumption of a six-hour drive corrective maintenance is more expensive than a weekly periodic maintenance strategy, but less frequent maintenance events than in the weekly scenario drive the costs back up again. The optimum in this case is the weekly maintenance strategy. For a journey of three hours, the most cost-effective strategy depends on the size of the PV plant. For the 1 and 10 MW_p plants, corrective and weekly periodic maintenance strategies are equally expensive. For the 100 MW_p plant, a periodic maintenance strategy is the most economical one. The advantage of periodic maintenance, combining multiple service team actions in one trip, becomes more important when the cost per trip rises. Driving six hours to a PV plant for every failure is more expensive than combining all repairs in a weekly maintenance event and accepting the resulting lost revenues. For a PV plant of the 100 MW_p size and with that the large number of components and failures, just three hours of driving time are enough for the plant operator to benefit from a periodic maintenance strategy.

The total costs for the 100 MW_p plant shown in Fig. 16 are divided into O&M and opportunity costs for 1, 3, and 6 hours of driving time. The opportunity costs for each maintenance strategy are almost constant for all driving times. A longer journey for the service team also means that a failed component remains in the failed state for the additional journey time. This downtime can result in further opportunity costs but it is too small to be detected in the figure. The O&M costs rise for every maintenance strategy with increasing driving time. It is most extreme in the corrective maintenance scenario, where the costs increase by around 14% for 3 hours and around 33% for 6 hours of driving time. A smaller maintenance frequency automatically leads to a smaller number of service team operations. For this reason, the increase in O&M costs is smaller for maintenance strategies with fewer maintenance events.

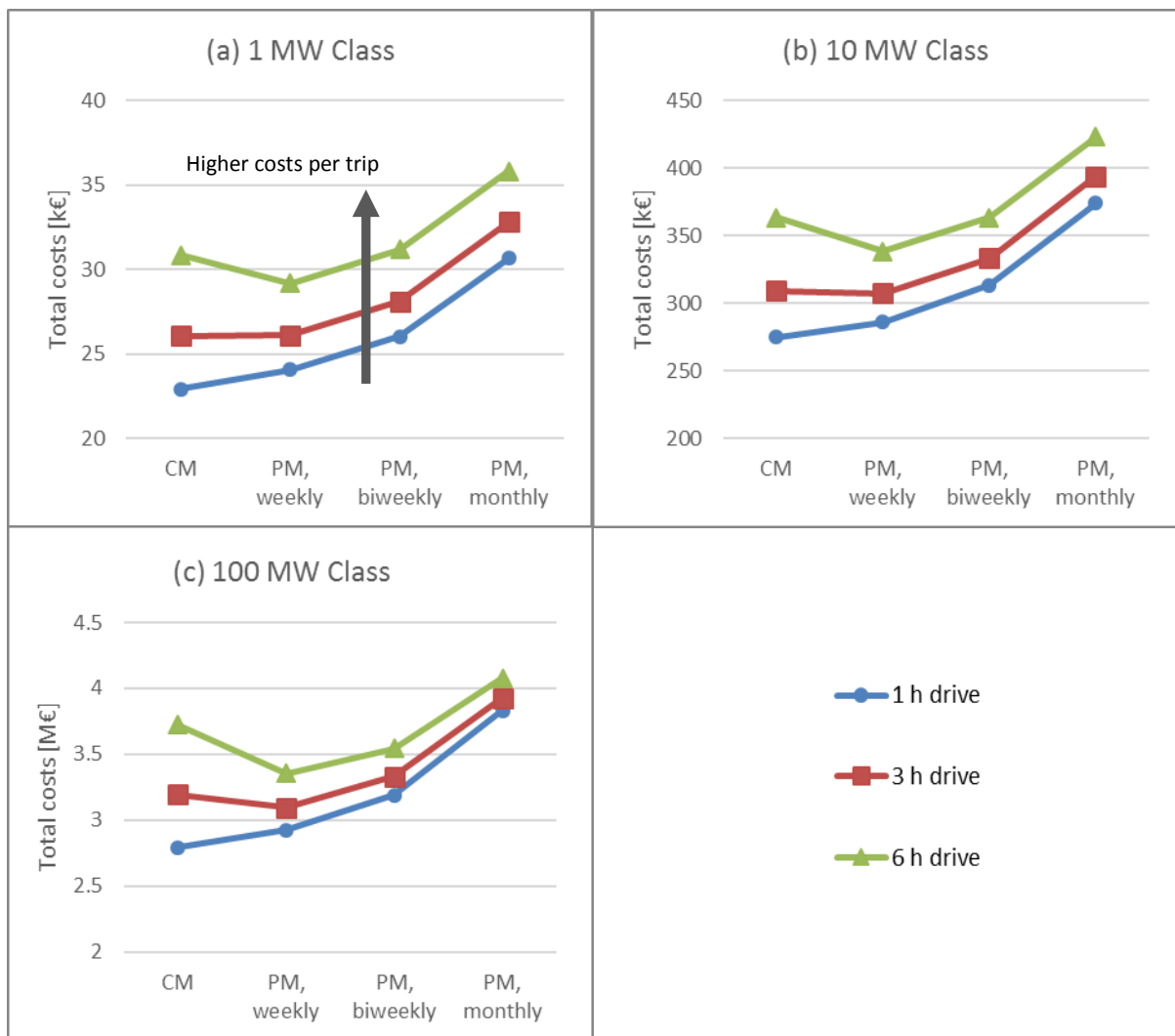


Fig. 15: Total costs for different maintenance strategies, varying the service team's driving time to the PV power plant (one way)

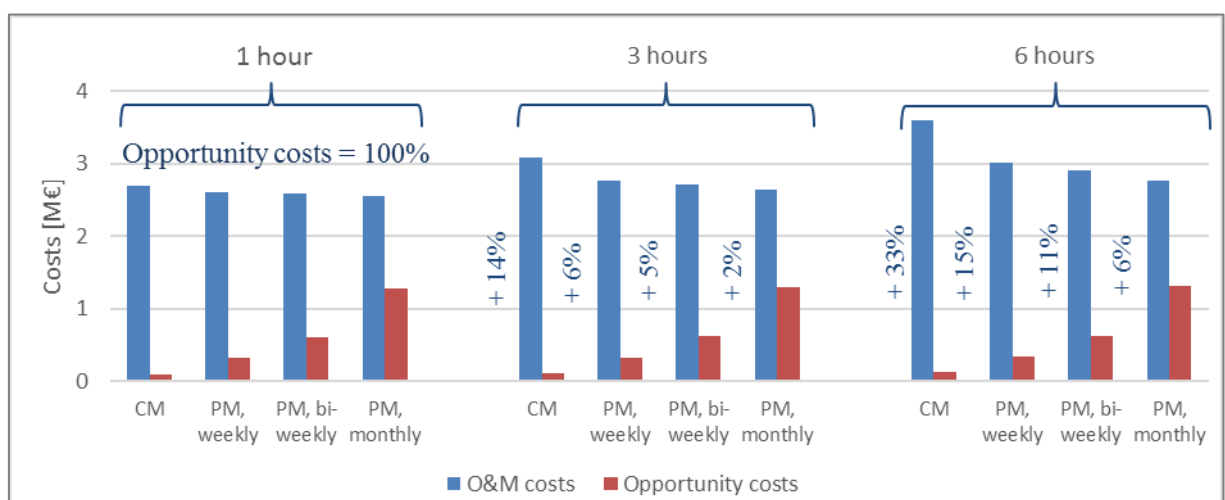


Fig. 16: O&M and opportunity costs for a 100 MW_p PV plant with different journey times for the service team

5.2.3 Spare Parts Inventory and Service Team Reaction Time

As stated in section 4.3, it is assumed for the corrective maintenance scenario that components are subject to lead times and that the service team reaction time is a random number between one and 24 hours. As the investigation above revealed that corrective maintenance is the most cost-efficient strategy for a PV plant, it should further be evaluated whether a spare parts inventory or a reduction of the reaction time can decrease total costs. Fig. 17 shows the potential savings that can be realized if the components' delivery lead times are assumed to be zero, i.e. there is a spare parts inventory on site.

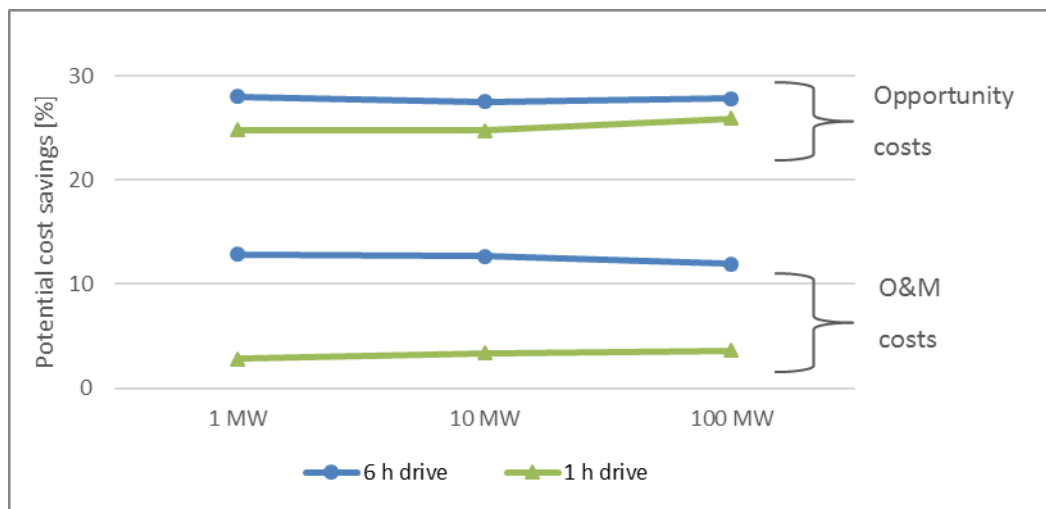


Fig. 17: Potential savings in O&M and opportunity cost because of spare parts inventory with 1- and 6-hour' driving time for the service team

Savings in O&M costs are achieved because spare parts are available right away on site. If components are not available right away, there is a service team operation one to 24 hours after failure and in case the failed component has a procurement lead time, there is another service team to install the component after the component's lead time has passed. This results in twice the amount of driving and higher labor costs. These additional costs can be avoided when every component is stored in a spare parts inventory on site. Savings are even higher when the service team's journey to the PV plant is six hours instead of one hour. Opportunity costs can be saved because downtime is reduced when components are renewed during the first service team's operation. If there was a lead time, the component and the affected array(s) would be offline until the second service team has installed the component after its lead time. The cost-saving potential on a percentage base is nearly constant for the PV plant sizes investigated.

The absolute potential cost savings for a 100 MW_p PV power plant with weather data from Bremerhaven and Swabian Albs are depicted in Fig. 18(a). The potential O&M cost savings are

in both cases about k€75, because the weather data do not have any influence on the service team. The potential opportunity cost savings increase from k€23.3 to 25.6 for the site with a higher global irradiation. The total savings are around k€100. It needs to be established whether these savings can justify the capital and operating costs associated with building and operating a spare parts inventory.

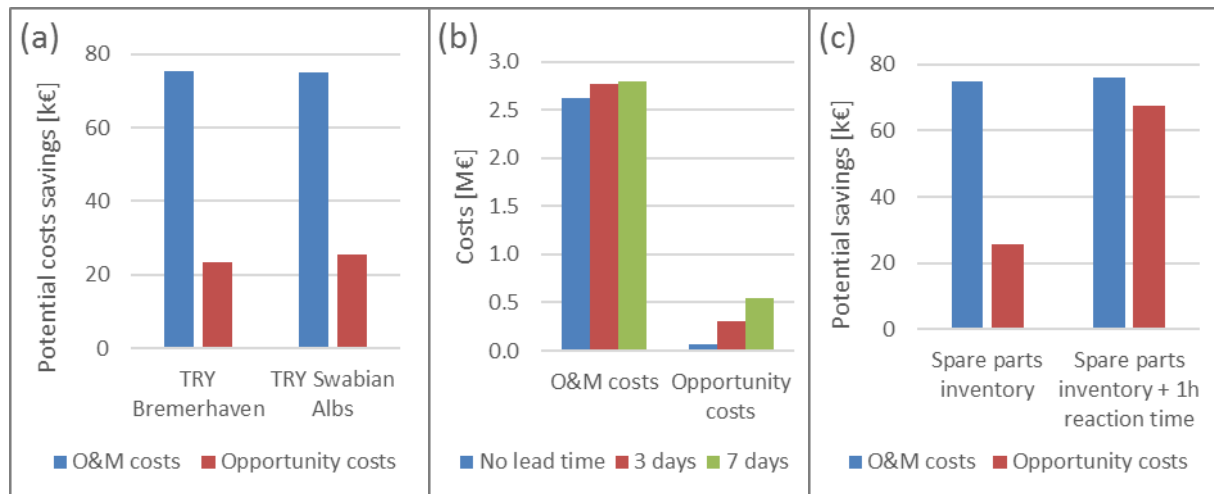


Fig. 18: For a 100 MW_p PV plant: (a) Potential costs savings because of a spare parts inventory with 1-hour' driving time and different weather data. (b) Comparison of O&M and opportunity costs with different lead times for all components. (c) Potential cost savings because of a spare parts inventory and a service team reaction time of 1 h after failure of a component and 1-hour' driving time

Fig. 18(b) depicts the development of the O&M costs as well as the opportunity costs if the lead time is increased to 3 and 7 days for all components. The O&M costs increase slightly due to the additional service team operations that have to be conducted if a lead time is assumed. The opportunity costs increase from k€70-308 if a lead time of three days is assumed. For a lead time of seven days, the opportunity costs amount to k€547, which is more than seven times the costs for the assumption of no lead time. This is due to the extended downtime of the arrays while the spare parts are being shipped. It has been shown that there is a cost-saving potential especially under the assumption of high lead times and higher global irradiation but the costs for an inventory need to be considered as well. The investment and operating costs for a spare parts inventory depend on multiple factors. It needs to be determined how much space is needed, how many components need to be held available, and how often they need to be restocked. The type of components that need to be stored possibly brings special requirements for the inventory, e.g. an air conditioning system to control the inventory's temperature and humidity.

There is also the possibility of reducing the service team reaction time. Fig. 18(c) shows the potential cost savings of a spare parts inventory combined with a service team reaction time of one hour. The spare parts inventory by itself leads to cost savings of 25%, but combined with the faster reaction time these amount to 65% of the opportunity costs in the base case. For a 100 MW_p PV plant that is around k€61. A faster reaction time can only be provided if enough personnel are available, which in turn leads to higher labor costs.

5.2.4 Inverter Reliability

The inverter is the component with the highest failure rate or the lowest mean time between failure. Fig. 19(a) displays the specific total costs for a 100 MW_p PV plant with different inverter failure rates. If the mean time between failure is assumed to be 20 years instead of 8 years, a cost reduction of about 50% can be achieved for all power/size classes considered. This reduction is due to lower opportunity costs, fewer O&M costs and lower costs for the acquisition of new inverters.

The comparison of the total costs for a corrective maintenance strategy and a weekly periodic maintenance strategy reveals that corrective maintenance is in both cases the most cost-effective choice (Fig. 19(b)). The difference in total costs of both strategies decreases slightly for an increased mean time between failure. This is due to fewer component failures in total and thereby lower opportunity costs through deferred maintenance in the periodic maintenance scenario. This trend is the same for higher power/size classes (see Appendix, Fig. A4). The inverter failure rate is an important factor for the reduction of the total costs but its influence on the maintenance strategy is relatively low.

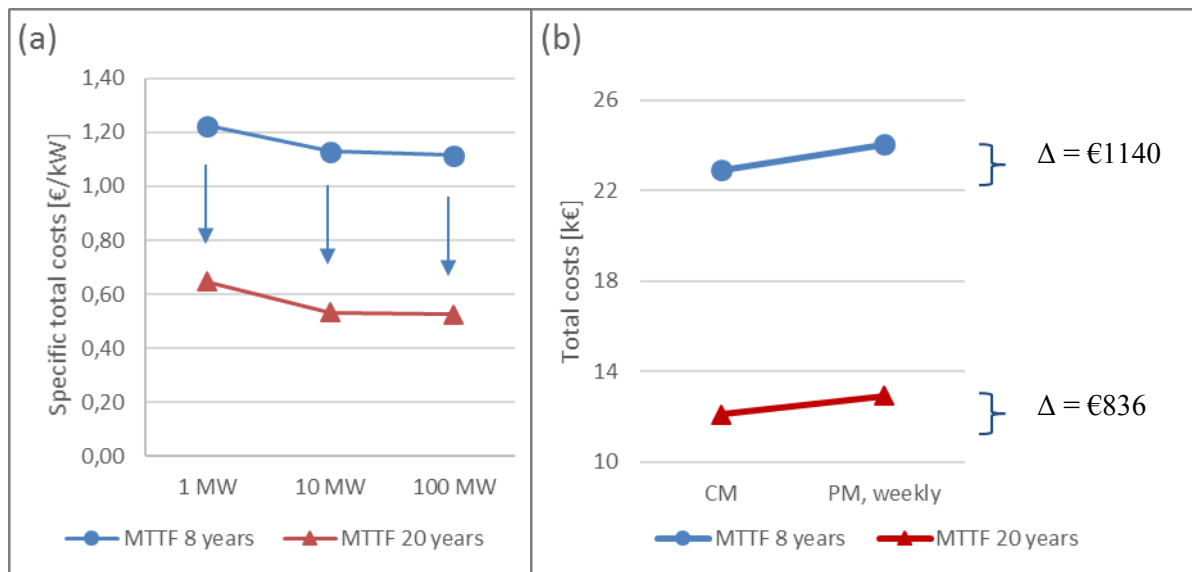


Fig. 19: For a 1 MW_p PV plant: (a) Cost reduction with a more reliable inverter and (b) comparison of corrective and weekly periodic maintenance strategies with a variation of the inverter mean time between failure

5.2.5 Electricity Spot Prices

The above-reported simulations have been performed under the assumption of constant electricity revenues of €89.4 per MWh. The following investigations have been conducted with the hourly mean European power exchange (EPEX) electricity spot price for the years 2011 to 2015.

All parameters except the electricity revenues are the same as the parameters used in the base scenarios in subsection 5.2.1. Unlike in the previous simulations, corrective maintenance is not the most cost-efficient strategy, but a weekly periodic maintenance is (Fig. 20(a)). A further decrease of the maintenance frequency to every two weeks leads to an increase in the total costs. The difference to the base cases is due to the decreased electricity revenues and hence lower opportunity costs. Averaged over an entire year, the EPEX electricity spot price is €39.18 per MWh, whereas the assumption for the previous simulations was a constant electricity revenue of €89.4 per MWh. A shift from corrective maintenance to weekly periodic maintenance leads to savings in O&M costs, and with the lower electricity revenues these savings are higher than the resulting opportunity costs because of lost sales. Fig. 20, plots (b) and (c) display the development of the opportunity and O&M costs when different maintenance strategies are applied.

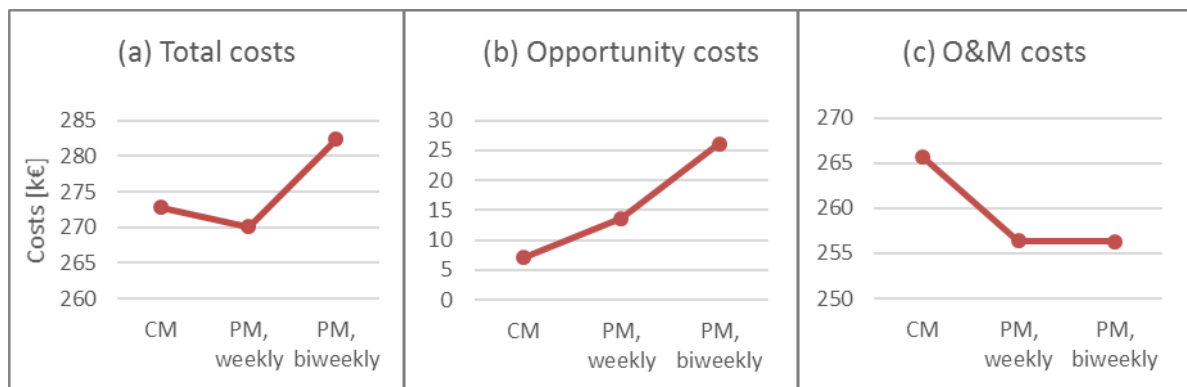


Fig. 20: Total, opportunity and O&M costs for different maintenance strategies with a variable electricity price for a 10 MW_p PV plant

Fig. 21 displays the opportunity costs for a weekly periodic maintenance strategy. The maintenance events are scheduled for Wednesday, Saturday, or Sunday at 8 am, 2 pm, or 8 pm. A clear trend for a difference in the day of the week cannot be identified. The time of day does have an impact on the resulting opportunity costs. Specifically, maintenance events scheduled for 8 am lead to the highest opportunity costs, followed by maintenance at 2 pm and the least costs emerge for maintenance events at 8 pm. This is accounted for by the behavior of the

electricity spot price and the global irradiation over the course of a day. The electricity price typically has a peak in the morning at around 8 am, followed by a trough in the afternoon and another peak in the evening at around 8 pm. The global irradiation has its peak at around noon, and sunrise and sunset depend on the time of year. At 8 am electricity prices are high and high opportunity costs emerge if maintenance is scheduled. Lower electricity prices lead to lower opportunity costs starting from 2 pm. The electricity price has another peak at around 8 pm but the global irradiation is very low and the sun possibly already down. This explains the relatively low opportunity costs occurring at 8 pm. The scheduling of maintenance events at specific times also has an impact on the O&M costs. Hourly labor costs are higher for work at night, Saturdays, or Sundays. The assumed hourly labor costs and hourly cost of driving has been increased 50% for work on Saturdays and 100% for work at night or Sundays. Fig. 22 depicts the total costs that result for the specifically scheduled maintenance events. Even though opportunity costs are the lowest for events at 8 pm the total costs are around the highest due to the additional cost for the night surcharge. The total costs for events at 8 am and 2 pm are higher from Wednesday to Saturday to Sunday. A 50% markup on hourly wages on Saturday and 100% on Sunday is the reason for this. The figure leads to the conclusion that the most cost-effective maintenance events are either Wednesdays at 8 am or 2 pm. Since the opportunity costs are lower at 2 pm and O&M costs are the same, the event at 2 pm creates slightly lower costs.

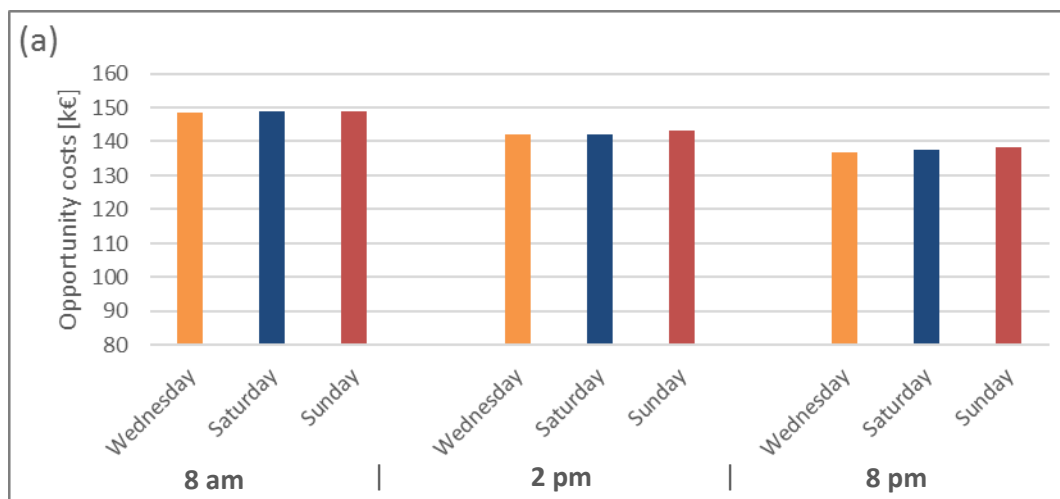


Fig. 21: Opportunity costs for a weekly periodic maintenance strategy scheduled at specific times, 100 MW_p PV plant

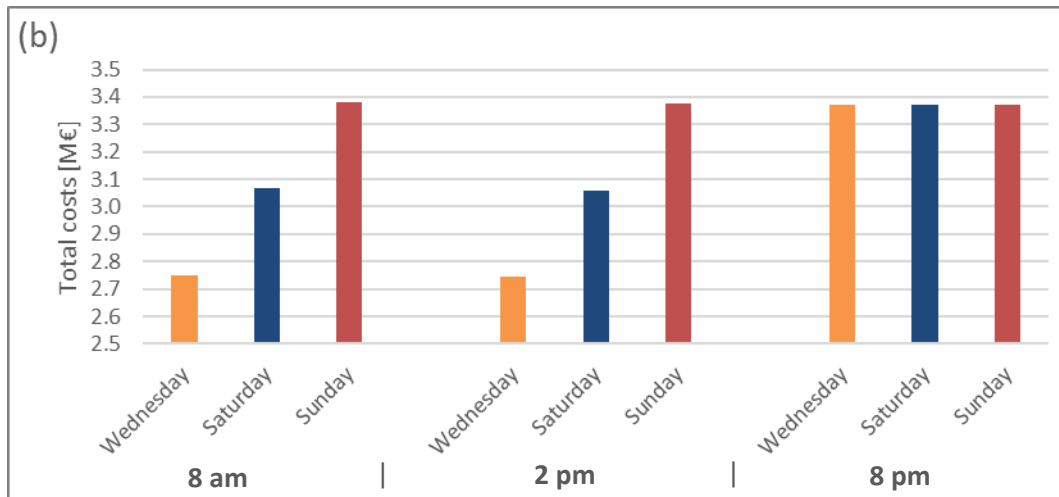


Fig. 22: Total costs for a weekly periodic maintenance strategy scheduled at specific times, 100 MW_p PV plant

5.3 Summary

The model developed is used to compare a corrective maintenance strategy, which implies the execution of corrective action right after the occurrence of a failure, with periodic maintenance strategies, which consist of periodically scheduled maintenance events, i.e. weekly, every two weeks and monthly. The simulations revealed that a lower maintenance frequency leads to an increase in unavailability because the corrective actions are deferred. The increased downtime of parts of the PV plant incurs higher opportunity costs and the additional costs can only be justified if they can be compensated by a decrease in O&M costs. This is not the case for the parameters chosen in the base scenarios, even though the O&M costs decrease slightly. Corrective maintenance is the most cost-effective strategy. This trend is the same for the 1, 10, and 100 MW_p PV power plants. An increase in the size of the PV plant leads to a decrease in the standard deviation of the total costs which is an important insight for risk management. The O&M costs are subject to scale effects and the costs per installed kW decrease with the PV plant size.

A variation of the service team driving time to the PV plant was also conducted. Since the service team is charged on an hourly basis, this means that the unit costs per service team operation rise. If the journey takes six hours instead of one hour, corrective maintenance is more expensive than a weekly periodic maintenance strategy. Multiple trips can be avoided when service team operations are not triggered by every failure but are scheduled on a regular basis. Accepting the resulting downtimes in a weekly maintenance scenario, and with that the emerg-

ing opportunity costs, results in lower total costs compared to the corrective maintenance strategy. This is the case for all three PV plant sizes. For a driving time of three hours, the size of the PV plant becomes a decisive factor. The 100 MW_p plant can be operated most cost-efficiently with the weekly maintenance strategy, whereas the 1 and 10 MW_p plants have about the same costs for the corrective and the weekly periodic maintenance strategy. This is due to the numerous components in the 100 MW_p plant. The total number of failures increases with the number of built-in components. For the 100 MW_p plant, the number is so high that it is not economical to send a service team out for every failure.

Further simulations have been conducted to evaluate the operation of a spare parts inventory. If corrective maintenance is chosen as a maintenance strategy, one should consider operating a spare parts inventory. It has been shown that a spare parts inventory decreases opportunity costs. For components with high lead times, for remote locations and for locations with higher global irradiation, the operation of an inventory is even more valuable. The emerging investment and operating costs for a spare parts inventory need to be determined with respect to the number and types of components that need to be held available.

An inverter mean time between failures of 20 instead of 8 years leads to a decrease of around 50% of the specific total costs since the inverter is the most vulnerable component. The decreasing effect is stronger for a weekly maintenance strategy than for a corrective strategy, because fewer service team operations in total lead to less deferred corrective action and thus lower emerging opportunity costs in the weekly maintenance scenario.

The application of the hourly mean European power exchange (EPEX) electricity spot price for the years 2011 to 2015 instead of a constant electricity price indicated that a lower electricity price leads to a shift from corrective maintenance to weekly periodic maintenance as the most cost-effective strategy. Due to the lowered electricity price, the opportunity costs in the weekly PM strategy are low enough to be compensated by the saved cost in light of reduced service team operations. An analysis of different days (Wednesday, Saturday, Sunday) and different times of the day (8 am, 2 pm, 8 pm) revealed that the opportunity costs are the lowest when service team operations are executed at 8 pm no matter what day of the week. Highest opportunity costs emerge for service team operations at 8 am due to high electricity prices and global irradiation in the morning. Service team operations at night or on the weekend are subject to night or weekend surcharges. A markup of 50% for Saturdays and 100% for Sundays and night operations on the hourly wages was assumed. The comparison of the resulting total costs revealed that maintenance events scheduled for Wednesdays at 8 am or 2 pm are the most cost-effective.

6 Conclusion and Outlook

With increasing numbers of photovoltaic plants worldwide and decreasing investment costs, the operating costs are becoming more significant for the viability of investments in photovoltaic power plants. The model developed in this study is a flexible tool to evaluate different maintenance strategies and their effects on O&M costs and opportunity costs. The O&M costs are determined by the present value of all maintenance actions. This includes costs associated with service team operations and costs stemming from the acquisition of new components. The opportunity costs are determined by the present value of all lost electricity revenues because of offline arrays. The model is implemented in a generic way so that the number, types, and characteristics of all components can be adjusted and different PV plant sizes simulated. This study focused on PV plants with a central inverter configuration and power outputs of 1, 10, and 100 MW_p, respectively. Arrays consist of a DC switch, a circuit breaker, and an inverter, and the arrays are connected to the grid by an AC switch and a circuit breaker. The focus of our study is put on the evaluation of possible actions after the occurrence of a failure. Periodically scheduled maintenance (PM) events as a form of deferred corrective maintenance are compared with the corrective maintenance (CM) which is carried out right away after the detection of failure. The investigated periodic maintenance frequencies are weekly, bi-weekly, and monthly.

The results show that with the parameter values assumed, corrective maintenance is the most cost-efficient strategy. By applying a periodic maintenance strategy, thereby deferring the corrective actions and reducing maintenance events, the O&M costs can be decreased slightly but not enough to compensate the emerging opportunity costs. Further simulations reveal that the *costs per service team trip* are an important factor in the evaluation of the maintenance strategy. When the costs per trip rise, for example, on account of an increase in the journey time of the service team, the ideal maintenance strategy shifts from CM towards PM events. Other possible reasons for increased costs per trip are higher personnel expenses or increased transport costs because of the need for a larger vehicle. Another decisive factor is the *size of the PV plant* because it determines the number of components and with that the total number of failures that occur over the lifetime of the plant. For a large PV plant with many components, a corrective maintenance strategy may lead to many service team operations that could be combined in periodic maintenance trips at a lower total cost. This cost advantage of PM does not necessarily apply to a smaller PV plant since it has a smaller number of total failures. The *electricity price* determines the costs associated with the unavailability of the PV plant. Our simulations show

that a decreasing electricity price leads to a shift to PM as the most cost-effective strategy, due to the savings in O&M costs and reduced consequences of PV plant unavailability. Investigations with a variable instead of a constant electricity price revealed that the selection of the *time of day and the day of the week* influences the resulting cost structure. Even though maintenance events in the night result in lower opportunity costs, the total costs rise compared to events scheduled in the day due to a night surcharge for the service team. The simulations show also that periodic maintenance was only viable if executed with a weekly frequency. A lower frequency results in higher total costs in all carried out simulations, due to the rising opportunity costs. The results imply that the ideal PM frequency decreases further when more extreme parameters are applied, i.e. lower electricity price, higher costs per service team trip, and larger PV plants.

A limitation of the proposed model is that the possibility to repair components is neglected. A failed component that can be repaired would be associated with smaller costs than a component that needs to be replaced. Further simulation should be conducted on a more detailed component level. However, this is restricted by the availability and accessibility of reliability data for the components. The same applies for the comparison between preventive and corrective maintenance strategies: Before these investigations can be conducted, the effect of preventive maintenance on the failure rates of components needs to be evaluated. A standardized collection of failure and maintenance data from a large number of photovoltaic plants is a key requisite for quantitative maintenance optimization and thus should be a focus of future investigations.

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Appendix

Appendix A.1: Failure rates for selected components

Table A1: Failure rates available in the literature for selected components

Component	Failure rate [10 ⁻⁶ Failures/h]	Cited in	Primary source
DC Switch	0.2	Zini et al. (2011), Ahadi et al. (2014), Perdue and Gottshalg (2015), Collins et al. (2010a)	US Department of Defense (1991)
	0.696	Colli (2015),	Power Systems Committee (1997)
Circuit Breaker	5.712	Zini et al. (2011), Ahadi et al. (2014), Perdue et al. (2015)	US Department of Defense (1991)
	0.4	Colli (2015)	Power Systems Committee (1997)
Inverter	40.29	Ahadi et al. (2014), Perdue et al. (2015)	Zini et al. (2011)
	175	Colli (2015)	Golnas (2013)
	86.4		Collins et al. (2010a)
	12.9		Cai et al. (2014)
AC Switch	0.034	Zini et al. (2011), Ahadi et al. (2014)	US Department of Defense (1991)
	0.696	Colli (2015)	Power System Committee (1997)

Appendix A.2: Unavailability probabilities

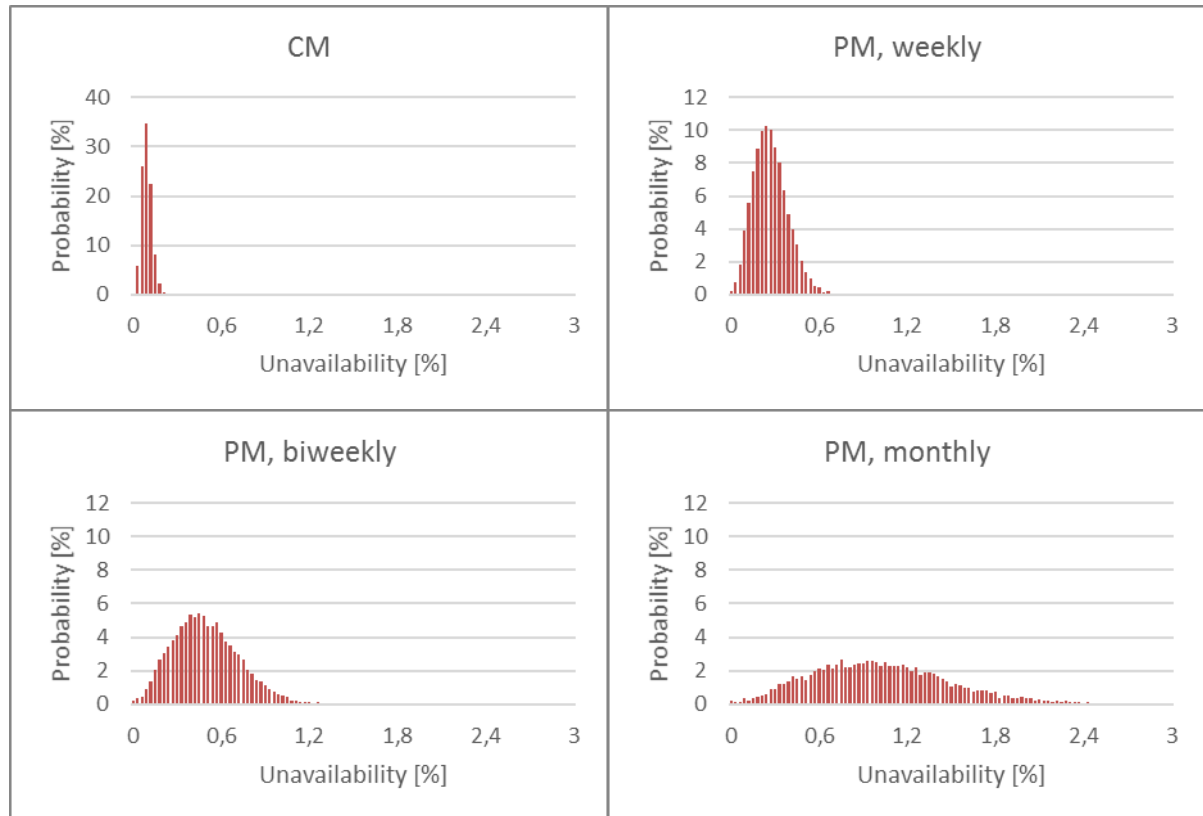


Fig. A1: Probability distribution of the unavailability of a 1 MWp PV plant

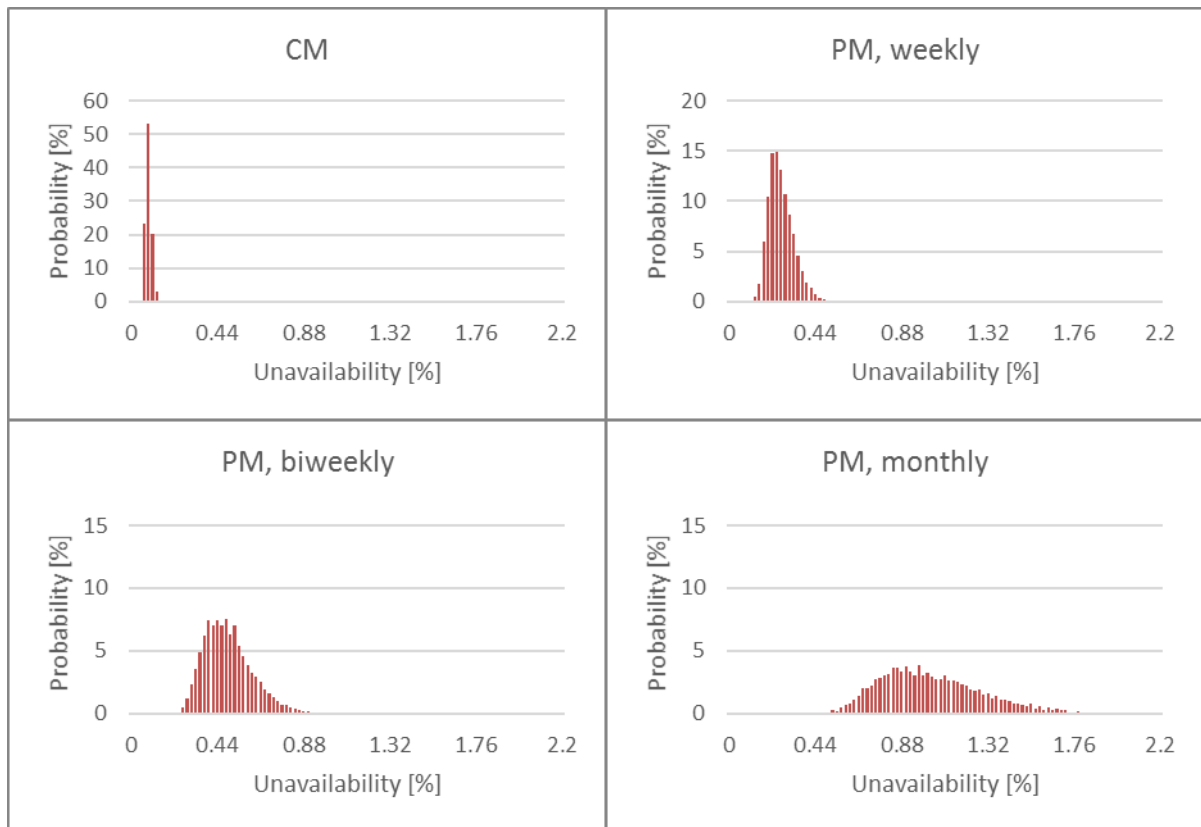


Fig. A2: Probability distribution of the unavailability of a 10 MW_p PV plant

Appendix A.3: Sensitivity analysis inverter mean time between failure

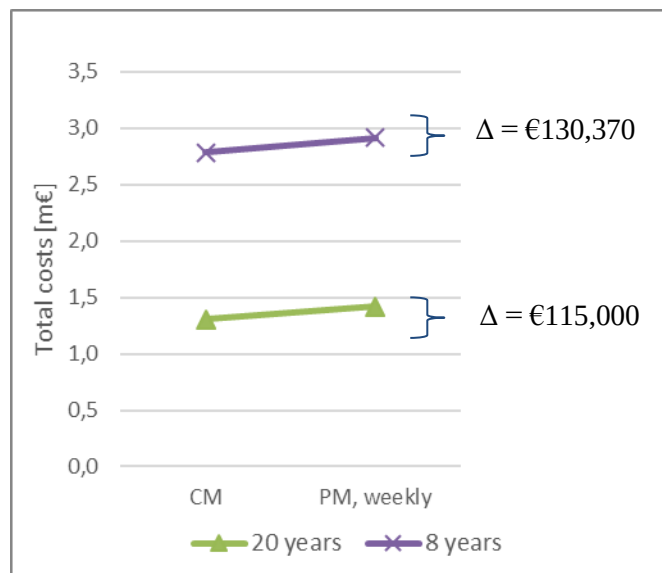


Fig. A3: Variation of the inverter mean time between failure for a 100 MW_p PV plant

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